



SENIOR THESIS IN MATHEMATICS

The Math of Moral Compromise Beyond Arrow's Impossibility

Author:
Federica Domecq
Lacroze

Advisor:
Dr. Vin de Silva

Submitted to Pomona College in Partial Fulfillment
of the Degree of Bachelor of Arts

August 21, 2026

Abstract

As a communal species, human survival has never been a solo endeavour but rather a collective one. At the crux of the human project lies the need to aggregate individual needs, taking a cacophony of human desires and harmonising them into single, actionable decisions. Beyond political convenience; we need agreement in order to build civilisations and navigate our differences.

Mathematically, the quest to aggregate desires remains fundamentally elusive. This body of work explores this question in a myriad of ways. Through historical attempts at creating fair voting mechanisms we navigate the inevitable flaws inherent in any mathematical structure.

We provide a mathematical explanation for these limitations by presenting Arrow's Impossibility theorem, demonstrating that any rank-order electoral system is doomed to fail, or result in dictatorship, unable to satisfy a set of minimal fairness axioms. Beyond the combinatorial proof, we introduce a topological approach implementing ultrafilters that allow us to better characterise Arrow's results. The ultrafilter framework allows for fascinating extensions: contemplating infinite electorates that allow for non-dictatorial outcomes.

We conclude by examining the current standing of democratic affairs in the 21st century and explore philosophical approaches to tackling democracy under impossibility.

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Chapter 1

Psephos, Ostraka, and Paradoxes: Social Aggregation Theory and An Early History of Voting

1.1 What is Social Aggregation Theory?

Our lives are driven by questions. What should I eat for breakfast? What time do I want to leave for class? Do I want to call my mother today? Questions lead to answers, which lead to actions. But seldom are our decisions made in isolation. We do not live in a vacuum, and any question posed to an individual can equally be posed to a collective.

When decisions belong to a group, something fundamental changes. Individual choice is simple: one mind, one preference, one answer. Collective choice is another matter entirely. With diversity of thought comes the potential for disagreement, and with disagreement comes a question that has occupied scholars for centuries — how do we take a cacophony of individual preferences and distil them into a single, coherent decision?

With varying degrees of importance, we face this question every single day, often believing that in the most ordinary of settings, a fair outcome is just a show of hands. Where should we go for dinner? Who should be our Commencement speaker? How should we spend the department's budget? And yet, the moment we try to answer it fairly, we run into trouble. Two

similar candidates split the vote, handing victory to a third that neither of their supporters wanted. A majority favourite emerges that almost everyone ranked last. No matter which rule we reach for, something feels wrong.

Coming up with a satisfactory rule turns out to be, not just difficult, but millennia-of-collective-human-effort difficult. It was not until 1951 that the economist Kenneth Arrow would formally explain why — proving, with mathematical precision, that the problem was not one of poor design but of fundamental impossibility. His monograph *Social Choice and Individual Values* (Arrow, 1950, 1951) earned him the Nobel Prize in Economics, making him the youngest Economics laureate at the time, at 51.

This work traces that story. In this Chapter, we begin by exploring the earliest attempts at fair collective decision-making and the flaws that undid them, continuing in Chapter 2 with the more sophisticated mechanisms that followed in the nineteenth and twentieth centuries. Chapter 3 presents a full proof of Arrow’s Impossibility Theorem, and Chapter 4 re-examines the result through the lens of ultrafilters, revealing a deeper structural explanation. We close in Chapter 5 by surveying the current landscape of electoral systems around the world and exploring what, if anything, can be done in the face of impossibility.

1.2 Early Forms of Preference Amalgamation

The premise of social aggregation theory is to amalgamate individual preferences into coherent social outcomes. As exemplified in the Introduction, we ask ourselves questions that would benefit from an optimal solution to this problem all the time.

Before turning to Arrow’s Theorem itself, it is valuable to get an understanding of the efforts that preceded this economist’s results. The novelty of his work can only be fully appreciated once we explore the many complex procedural paradigms scholars have designed in hopes of coming up with an optimal solution. The main goal of this chapter is to convey how hard extracting consensus really is by exploring historical attempts at solving the dilemma. Prior to Arrow’s work, every attempt revealed different procedural inefficiencies and contradictions. These “failed” attempts convey that individual rationale doesn’t really scale easily.

Like many other mechanisms, the notion of voting was first documented in the Classical World. Many of the electoral systems we use today, were

first conceived in Ancient Greece and Rome.

1.2.1 Plurality, The Romans & The Greeks

The earliest recorded “structure” of voting can be found in the heart of the Spartan political system, embodied in the Great Rhetra (an allegedly oral constitution given by the Delphic Oracle to the Spartan lawgiver Lycurgus) which was believed to have originated in the 7th century BC Girard (2010). The governing body of this society was constructed with four components: the Diarchy, the Gerousia, the Ephorate, and the Apella.

The Gerousia, which was the *Council of Elders* and was responsible for drafting and initiating legislation, consisted of 30 members who were elected by “acclamation”. A form of voting whereby candidates were led past the assembly, and the winner was determined by the loudness of the shouting recorded by the observers. While seemingly outrageous, this mechanism established the principle that individual opinions could be aggregated into collective decisions. We might understand this mechanism as a form of vote by *Plurality*.

Definition 1.1 (Plurality). Let $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$ be a finite set of alternatives and $V = \{1, 2, \dots, n\}$ be a finite set of voters. Each voter $i \in V$, denotes their highest preference $p_i \in \mathcal{A}$. The plurality score for any alternative $x \in \mathcal{A}$ is the number of voters who placed x as their top preference, that is:

$$S(x) = \sum_{i=1}^n \mathbf{1}(p_i = x)$$

Where $\mathbf{1}$ is the indicator function.

$$\mathbf{1}(p_i = x) = \begin{cases} 1 & \text{if voter } i \text{ places } x \text{ as their top choice} \\ 0 & \text{otherwise} \end{cases}$$

In the case of *Plurality*, the winner $w \in \mathcal{A}$, is the alternative that maximises the score function. That is,

$$\text{Winner} = \{x \in \mathcal{A} \mid S(x) \geq S(y) \text{ for all } y \in \mathcal{A}\}$$

Other Greeks, particularly Athenians, performed a similar democratic procedure, which was more precise and quantifiable. The assembly mostly

used a *show of hands* (cheirotonia), to decide legislative bills, declarations of war, and the election of certain officials. This procedure was most commonly used with majority rule, Garner (1993).

Remark 1.2. While the plurality winner described in 1.1 is the alternative maximising the score function $S(x)$, a **majority** winner is an alternative w such that $S(w) > n/2$. Notice that a majority winner is always a plurality winner, but a plurality winner is not necessarily a majority winner.

It is worthy of note that the assembly was aware of the potential for corruption in public voting; once they brought “show of hands” to larger groups, the task of estimating the number of votes for each candidate was not trivial. For delicate cases and for the process of ostracism – an exercise where citizens could nominate an individual who they deemed a threat to the state for exile– cast votes on *ostraka* (pottery shards). If at least 6000 votes were cast, the individual who got the most votes by plurality was banished for a decade, Salamanca (2023). This is a clear example of social aggregation actually used for negative selection.

1.2.2 The Flaws in Plurality

One could dedicate an entire book to the history and iterations of Plurality voting. In this section we are going to limit ourselves to describing the limitations inherent in this form of preference aggregation.

The Spoiler Effect

Plurality suffers from the Spoiler Effect.

Definition 1.3 (The Spoiler Effect). The spoiler effect occurs when the addition or removal of a non-winning candidate (C) in an election alters the outcome such that a different candidate (A instead of B) wins.

One could imagine a scenario where two candidates, candidate A and candidate B which are relatively similar run against candidate C . The existence of two similar candidates can cause the majority of the vote to split, allowing C to win even if a vast majority of the voters prefer either A or B . A specific example of this phenomenon follows.

Example 1.4. In the 2000 US Presidential elections, voters observed a clear instance of the Spoiler Effect. Republican George W. Bush won the state

of Florida over Democrat Albert Al Gore by a narrow margin of 537 votes out of close to 6 million, by 0.00895%. In the electoral ballot, was the Green Party nominee, Ralph Nader, who was the first choice of 97,488 Floridians. The distribution of votes can be seen in Figure 1.1a.

It is not hard to imagine, that Nader’s voters would have preferred Al Gore to win over Bush. Had Nader not been a choice, we might have seen a distribution of choices like that presented in Figure 1.1b.

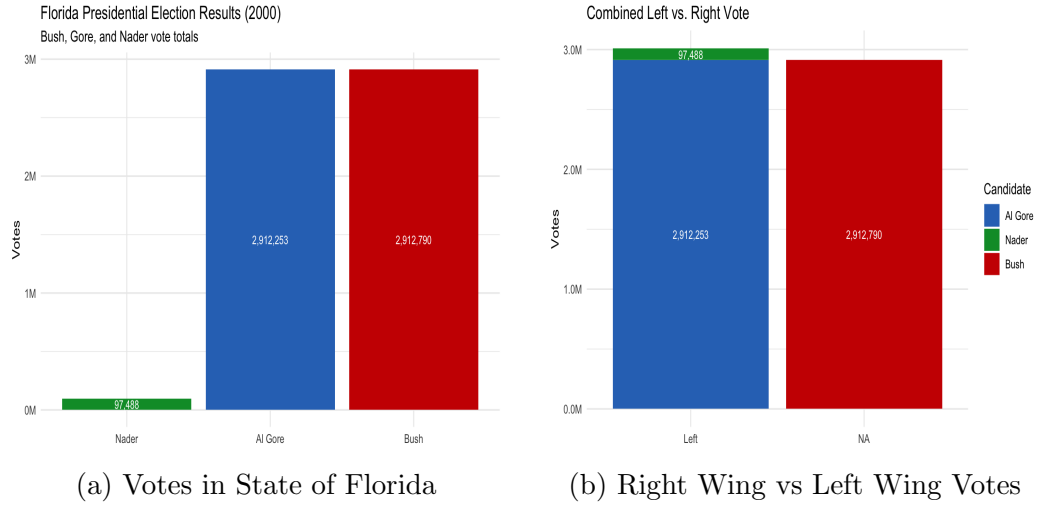


Figure 1.1: Outcomes in 2000 US Presidential Election

The Spoiler Effect is the result of the violation of the Independence of Irrelevant alternatives, a condition posed by Kenneth Arrow which we describe and explore further in Section 3.2.

Failure of Condorcet Criterion

Plurality also violates the Condorcet Criterion.

Definition 1.5 (Condorcet Criterion). If there happens to be a choice in a pairwise comparison of alternatives that is preferred by the voters to every other choice, then that choice should be the winner of an election.

Example 1.6. One can imagine an election where we have 10 voters whose preference over the following alternatives, $\mathcal{A} = \{P, Q, R, S, T\}$, have to be

aggregated. One possible outcome of the election can be visualised in Figure 1.2.

Under Plurality, P is the winner because it has the highest score of $S(P) = 3$. If we look at the 5th choice column, however, we realised that 70% of voters ranked P last! Using Plurality, 70% of voters are left to deal with their last-ranked choice.

Furthermore, in any pairwise comparison between P and the other alternatives, P will lose miserably. For example, voters P_4 through P_{10} prefer R over P , which implies that in a pairwise comparison, R will clearly defeat P (7 : 3).

Given that P loses to other candidates in pairwise comparisons, it follows that P should not be the winner under Condorcet's criterion. In fact, in this

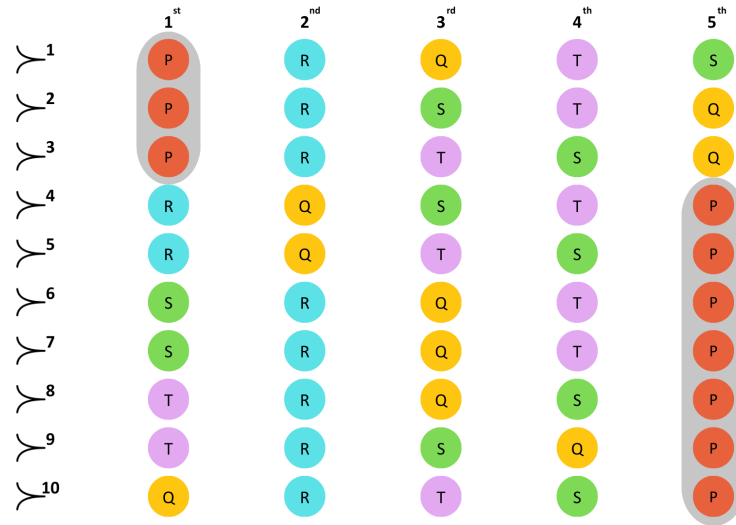


Figure 1.2: This represents our 10 voters' full ranking. Each column of dots represents the appropriately ranked options.

example, R should be the Condorcet winner given that its ranked 2nd by P 's voters (and we already indicated how that pairwise comparison is won by R) and it's ranked first or second by everybody else.

Lack of Majority

Potentially, one of the more intuitive limitations of the Plurality rule, is that it does not require a candidate to present absolute majority to win. After all,

seldom does this sort of result occur. As illustrated in Example 1.6, while the Plurality winner maximises the score function $S(x)$, they can often do so while presenting a very small percentage of the first choice votes when the voters are presented with sufficient alternatives.

Under Plurality, alternatives can win even if a vast majority of the voters prefer another alternative. Again, this example is illustrated in Example 1.6 where R should be the clear winner.

Strategic Voting

A direct consequence of the Spoiler Effect is the issue of strategic (or tactical) voting. In this instance, individuals may try to game the voting mechanism to ensure that they are not left with the “greater of two evils”. Under Plurality, voters may abandon their true first choice for a more “electable” alternative in order to prevent the consequences of the Spoiler Effect. It follows that this voting mechanism often fails to capture the sincere preferences of the electorate, an argument that follows the Gibbard-Satterthwaite Theorem Benoît (2000).

Theorem 1.7 (Gibbard-Satterthwaite, 1975). *For deterministic, ordinal, single-winner voting elections with $\mathcal{A} \geq 3$ alternatives, the only strategy-proof, voting rule is that of a dictatorship.*

1.3 Voting Theory in the Medieval Era

1.3.1 Ramon Llull & Pairwise Comparison

Mallorcan Catholic philosopher, Ramon Llull has most recently been credited with the earliest sophisticated contemplations of ranked voting by Hägele and Pukelsheim (2001).

Llull hoped to revolutionise the often-corrupt process of selecting leaders within the Church. By creating a systematic and unbiased approach to elections, he hoped that his *Ars* would help align the Church’s elections with divinity, reducing personal bias, finding consensus in undeniable binary choices and creating a mechanism that would yield virtuous candidates. His system, described in *De Arte Electionis* outlined a round-robin tournament predating Marquis de Condorcet by five centuries.

Definition 1.8 (Llull's Pairwise Comparison Rule). Let $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$ be a finite set of alternatives and $V = \{1, 2, \dots, n\}$ be a finite set of voters. For every distinct pair of alternatives $\{x, y\} \subseteq \mathcal{A}$, we define a pairwise competition.

We compute the number of voters who prefer x over y ,

$$T(x > y) = |\{i \in V : x \succ^i y\}|$$

The winner of a pairwise contest between x and y is given by:

$$W(x, y) = \begin{cases} 1 & \text{if } T(x, y) > T(y, x) \\ 0 & \text{otherwise} \end{cases}$$

The Llull score given to an alternative $x \in \mathcal{A}$ is the sum of its victories across all pairwise comparisons with alternatives $y \in \mathcal{A}$:

$$\text{Llull}(x) = \sum_{y \in \mathcal{A}, y \neq x} W(x, y)$$

The winner of an election is the alternative with the highest Llull score.

Example 1.9. Consider an election within a small monastic order with three candidates, $\mathcal{A} = \{a, b, c\}$.

Llull's method requires us to evaluate all $\binom{3}{2} = 3$ possible pairwise competitions.

Suppose the aggregate results of these contests are as follows:

- (1) a vs b : a receives 6 votes, b receives 4 votes. (a wins)
- (2) b vs c : b receives 7 votes, c receives 3 votes. (b wins)
- (3) a vs c : a receives 8 votes, c receives 2 votes. (a wins)

We calculate the Llull scores $\text{Llull}(x)$ by summing the victories:

- $\text{Llull}(a) = W(a, b) + W(a, c) = 1 + 1 = 2$
- $\text{Llull}(b) = W(b, a) + W(b, c) = 0 + 1 = 1$
- $\text{Llull}(c) = W(c, a) + W(c, b) = 0 + 0 = 0$

In this case, candidate a is the unique Llull winner with 2 pairwise victories. Notice that a is also a Condorcet winner, as it remains undefeated in all possible binary matchups.

1.3.2 Llull's Pairwise Flaws

While seemingly logical, pairwise comparison is not without its flaws.

Computational Complexity

The most practical drawback at the time was the computational complexity of this exercise. With k alternatives, Llull's method requires us to compute $\binom{k}{2} = \frac{k(k-1)}{2}$ pairwise comparisons. Once we consider 20 alternatives, we have to compute 190 comparisons. In a medieval setting, where votes were cast by hand, hosting 200 separate elections and computing the pairwise comparison results was practically intractable.

Condorcet Irrationality

While Llull was the first to recognise that voter preferences could be irrational and non-transitive, he failed to account for this in his preference aggregation mechanism. He discovered what is commonly known as Condorcet's Paradox but his work was lost until the 21st century, which explains why the Paradox is often attributed to the Marquis de Condorcet who only made this discovery in the late 18th century.

Definition 1.10 (Transitive Preferences). A ranking of preferences is considered transitive if it presents a consistent hierarchical order with a clear favourite, followed by alternatives strictly ranked lower. Formally for any three or more alternatives, $\mathcal{A} = \{x, y, z\}$, if a voter prefers x over y , and y over z , then they must prefer x over z .

We see Condorcet's Paradox in scenarios where even though all voters have transitive rankings, the aggregation mechanism presents an irrational intransitive result. Examples of Condorcet's Paradox are known as cyclic ties.

Example 1.11 (Cyclic Tie Using Llull's Social Aggregation Mechanism). Suppose an election within a small monastic order presents three candidates, $\mathcal{A} = \{a, b, c\}$. There is a small assembly of three voters $|V| = 3$. Instead of having voters provide pairwise rankings, we request they fully rank their alternatives from favourite (1st choice) to last, to ensure that their preferences are transitive. As such, they provide:

	$\succ^1$	$\succ^2$	$\succ^3$
1 st	a	b	c
2 nd	b	c	a
3 rd	c	a	b

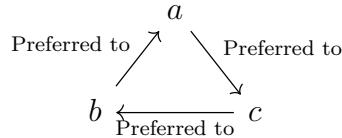
Llull's method requires us to evaluate all $\binom{3}{2} = 3$ possible pairwise competitions. We perform the comparisons here:

- (1) a vs. b : a receives 2 votes, b receives 1 votes. (a wins)
- (2) b vs. c : b receives 2 votes, c receives 1 votes. (b wins)
- (3) a vs. c : a receives 1 votes, c receives 2 votes. (c wins)

We calculate the Llull scores $\text{Llull}(x)$ by summing the victories:

- $\text{Llull}(a) = W(a, b) + W(a, c) = 1 + 0 = 1$
- $\text{Llull}(b) = W(b, a) + W(b, c) = 0 + 1 = 1$
- $\text{Llull}(c) = W(c, a) + W(c, b) = 1 + 0 = 1$

When we aggregate these results, we find a collective preference:



The diagram above captures the resulting societal preference: a is preferred to b by majority, b is preferred to c by majority, however c is preferred to a by majority. Instead of producing a linear hierarchy, the aggregation cycles back on itself, leaving no alternative that can be declared the societal favourite. We find a *cyclic tie* violating transitivity.

1.3.3 Nicholas of Cusa & Positional Ranking

As highlighted by Plurality and Llull's pairwise paradigm, there may be value in having a full understanding of the voter's complete set of preferences.

In the scheme of Plurality, having these rankings allow us to hypothesise just how happy (or unhappy) the group of voters may be with the outcome. In Example 1.6 specifically, we were able to see that with the Plurality outcome a majority of voters would be discontent – over 50% of voters ranked P , the winner, in last. Similarly, the example allowed us to see whether the results violated the Condorcet Criterion.

In the case of Llull, requiring a voter's full ranked set of preferences, can indirectly prevent a mechanism whereby we have to run $\binom{k}{2} = \frac{k(k-1)}{2}$ round-robin tournament, given that it is mathematically equivalent to infer these pairwise preferences from a fully ranked set, such as in Example 1.11.

It was Nicholas of Cusa, a German Catholic bishop, who, in realising that pairwise comparison could fail to produce a resolute winner, proposed a mechanism which leveraged completely ranked ballots. His work was a proposed system for electing a King for the Holy Roman Empire, discarding binary comparisons in favour of a scoring method more commonly known as Borda Count (as Jean Charles de Borda re-devised the system during the French Revolution).

Definition 1.12 (Borda Count). Let $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$ be a finite set of k alternatives and $V = \{1, 2, \dots, n\}$ be a finite set of voters. Each voter $i \in V$ provides a strict linear ordering of all alternatives.

The Borda score for an alternative $x \in \mathcal{A}$ from a single voter $i \in V$ is determined by the number of alternatives ranked below it. Formally, the score individual i assigns to x , $s^i(x)$, is:

$$s^i(x) = |\{y \in \mathcal{A} \text{ such that } x >^i y\}|$$

The total Borda score for x , is the sum of the individual scores $s^i(x)$ across all voters.

$$S(x) = \sum_{i=1}^{|V|} s_i(x)$$

The winner is the alternative x that maximises this score:

$$\text{Winner} = \{x \in \mathcal{A} \mid S(x) \geq S(y) \text{ for all } y \in \mathcal{A}\}$$

Example 1.13. Suppose we have a set of 3 alternatives, $\mathcal{A} = \{a, b, c\}$ and we have a population of five voters, $|V| = 5$. They fully rank their alternatives from most preferred to least preferred, and these ballots are displayed in Table 1.1a with their corresponding point assignments displayed in 1.1b.

	$\succ^1$	$\succ^2$	$\succ^3$	$\succ^4$	$\succ^5$
1 st	a	a	b	b	c
2 nd	b	b	c	c	a
3 rd	c	c	a	a	b

(a) Ranking Orders. Each column corresponds to an individual voter's ballot; each row corresponds to a ranking position.

	$\succ^1$	$\succ^2$	$\succ^3$	$\succ^4$	$\succ^5$
3 pts	a	a	b	b	c
2 pts	b	b	c	c	a
1 pts	c	c	a	a	b

(b) Point Assignments. Each column corresponds to an individual voter; each row denotes how many points an alternative is assigned.

Once we have the point assignments, we can compute the Borda Count of these alternatives by tallying the total points assigned to each alternative. This process can be observed in Table 1.2, giving **b as the winner**.

	a	b	c
$\succ^1$	3	2	1
$\succ^2$	3	2	1
$\succ^3$	1	3	2
$\succ^4$	1	3	2
$\succ^5$	2	1	3
Total	10	11	9

Table 1.2: Borda Count Calculation

1.3.4 Frailties of Borda Counting

Unsurprisingly, Borda Count is not without its own inherent limitations.

Strategic Manipulation

Namely, in 1814 Pierre-Simon Laplace, one of Borda Count's strongest critics, argued that this ranking mechanism was vulnerable to strategic manipulation, whereby intelligent voters would "place in the last rank the candidate

most formidable to that one whom they prefer”, this indirectly gives “too great an advantage to candidates of mediocre merit” who are unlikely to receive such harsh “punishment”, Laplace (1814).

While never withdrawing his criticism of Borda Count, Laplace unintentionally gave Borda Count its strongest mathematical justification by finding that in a Bayesian contemplation, this assignment of weights was indirectly the maximum likelihood estimator. Interested readers can find an outline of his argument and proof in Appendix A.

Once again, we can reference the Gibbard–Satterthwaite Theorem (Benoît, 2000), which formalises the fact that the only mechanism which captures the sincere preferences of the electorate is a dictatorial one.

Candidate Interference

One of the most evident shortcomings of this method is its susceptibility to candidate interference, whereby the introduction or removal of an irrelevant candidate can modify the relative ranking of two other candidates, even if the preferences between the two haven’t changed.

Example 1.14. Take the same set up and preferences given in Example 1.13. Say, for whatever reason, that we no longer have alternative c as an option, leaving us with $\mathcal{A} = \{a, b\}$. The relative ordering given by the voters should not change. Maintaining their relative ordering would give us:

	$>^1$	$>^2$	$>^3$	$>^4$	$>^5$
3 pts	a	a	b	b	$\underline{c}$
2 pts	b	b	$\underline{c}$	$\underline{c}$	a
1 pts	$\underline{c}$	$\underline{c}$	a	a	b

(a) Items to remove

	$>^1$	$>^2$	$>^3$	$>^4$	$>^5$
2 pts	a	a	b	b	a
1 pts	b	b	a	a	b

(b) Point Assignment with Same Relative Ordering

The new Borda Count results,

	a	b	c
Total	8	7	×

With the same relative ordering between a and b , now a wins, showing how c influenced the result indirectly.

Borda's paradigm similarly fails Condorcet's criterion, presented under Llull's Pairwise comparison's limitations in Section 1.3.2.

1.4 Liberté, Égalité, Absurdité

In the late eighteenth century, France underwent radical transformation as its lower classes saw a period of Enlightenment catalysing a revolution against the standing monarchy and the idea of the divine rights of kings.

The thinkers of the time, philosophers and mathematicians, anticipated the most relevant question — if the nation is left to the “will of the people”, how exactly can we govern?

The quest for rational mechanisms to define democracy was birthed, and the term *mathématique sociale* (social mathematics) was coined. The goal was to find mechanisms replacing aristocratic decree with objective and egalitarian processes. This question drew the interest of two of the most famous figures in voting theory: Jean-Charles de Borda and the Marquis de Condorcet. It is worth mentioning that the most pressing motivator for these results was not the French government but the French Academy of Sciences.

1.4.1 Jean-Charles de Borda Versus The Marquis de Condorcet

In 1770, Borda presented his positional scoring system — which was equivalent to the aforementioned voting mechanism implemented by Cusa (Section 1.3.3) — arguing that Plurality was fundamentally flawed and often elected candidates who were broadly disliked. Though flawed, Borda Count was eventually adopted by the Academy in 1784.

The Marquis de Condorcet, however, a fellow academician, fiercely criticised Borda's method.

He explored alternative mechanisms of Social Aggregation, more concretely, pairwise comparison mechanisms such as that implemented by Llull and explored in Section 1.3.2. In 1785, Condorcet published his *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix*. In it, he mathematically formalised the paradox that Ramon Llull

had stumbled upon centuries earlier, proving that majoritarian preferences, when considered in pairwise manners, could be cyclic. Furthermore, Condorcet demonstrated that Borda's method could easily fail to elect a candidate who would beat every other candidate in a one-on-one matchup, that is, that it fails the Condorcet Criterion.

Ultimately, this period was defined by absurdity. Borda and Condorcet, two brilliant minds equipped with the highest mathematical rigour of their time, attempted to construct an infallible democratic system. Inadvertently, their rivalry proved that rational collective decision-making seemed to be inherently paradoxical as neither man could fix the other's system without inheriting its defects.

Their work laid the groundwork for the next century of mathematicians. In hindsight, we come to the conclusion that their work was not a failure but a symptom of something deeper. The recurring pathologies would reappear in every subsequent attempt to redesign the perfect voting mechanism. It would take until 1951, through Arrow's Impossibility Theorem, to understand why.

Chapter 2

Refined Procedures and Recurring Failures

Attempts to reconcile the issues presented by existing voting theory did not end with the resolution of the French Revolutionary period. If anything, the notoriety of the individuals researching this question drew more attention to the topic. The nineteenth century saw many more mathematicians tackling this problem by presenting mechanisms designed to learn from their predecessors' flaws.

2.1 Single Transferable Vote

One of the methods presented in this time was that of Single Transferable Vote, often known as Proportional-Ranked Choice Voting. While British barrister Thomas Hare is traditionally attributed the credits for the creation of this multi-winner electoral system in 1857, it was first proposed by English mathematician Thomas Wright Hill in 1819, Tideman (1995).

Far from trying to amend the state of democracy, Hill was a British school teacher who was motivated to tackle the question of voting systems when trying to fairly elect a committee for the Society for Literary and Scientific Improvement of Birmingham. Hill recognised that under Plurality many of the votes provide no contribution to electoral outcomes. By creating a new system, he hoped that votes wouldn't be wasted on already-winning candidates or hopeless ones — that minority votes could in fact become useful. His goal was to create a system in which votes would transfer to

where they actually mattered. As such, he came up with the voting system described next.

Definition 2.1 (Single Transferable Vote). Let $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$ be a finite set of alternatives and $V = [n]$ be our finite set of voters. Each $i \in V$ provides a strict linear ordering $>^i$ of all the given alternatives.

We define a minimum majority threshold:

$$\text{Maj} := \left\lfloor \frac{n}{2} \right\rfloor + 1$$

We may have several rounds of “counting”, let $\mathcal{A}^{(r)} \subseteq \mathcal{A}$ denote the set of remaining alternatives in a given round, with $\mathcal{A}^{(1)} = \mathcal{A}$. Each voter i ’s active preference in round r is their first-ranked alternative in $\mathcal{A}^{(r)}$, under the restriction of their original rankings to the set of remaining alternatives $\mathcal{A}^{(r)}$. We keep a count of the number of votes each alternative $x \in \mathcal{A}^{(r)}$ gets in round r :

$$S^{(r)}(x) = |\{i \in V : x \text{ is voter } i\text{'s active preference in round } r\}|$$

The procedure then iterates as follows:

- (1) (Winner, completed election) If $S^{(r)}(x) \geq \text{Maj}$ for some $x \in \mathcal{A}^{(r)}$, then said x is declared the winner.
- (2) (Elimination) Otherwise, the alternative with the lowest tally is eliminated:

$$\mathcal{A}^{(r+1)} = \mathcal{A}^{(r)} \setminus \left\{ \arg \min_{x \in \mathcal{A}^{(r)}} S^{(r)}(x) \right\}$$

and the tallies are recomputed under the restricted ordering that is given by removing the least popular option. This process repeats until a winner is declared.

Remark 2.2. The definition above, 2.1 presents Single Transferable Vote in a single-winner formulation, which suffices for the purposes of the arguments presented below. Note that said single-winner formulation is often referred to as a *Runoff election*.

Furthermore, there are definitions for Single Transferable Vote for varying numbers, x , of seats to be assigned. In said setting, the STV procedure terminates once x candidates can be elected.

Another goal presented by this voting system is that of sincere preference expression. As postulated by Hare, (Hare, 1859), this system aims to ensure that voters can express their true preference without fear of wasting their vote or inadvertently helping their least favourite candidate.

Example 2.3. Consider an election with three alternatives $\mathcal{A} = \{a, b, c\}$ and nine voters $|V| = 9$. The individual ballots are in Table 2.1.

	$\succ^1$	$\succ^2$	$\succ^3$	$\succ^4$	$\succ^5$	$\succ^6$	$\succ^7$	$\succ^8$	$\succ^9$
1 st	a	a	a	b	b	b	b	c	c
2 nd	b	b	c	a	a	c	c	a	a
3 rd	c	c	b	c	c	a	a	b	b

Table 2.1: Rankings given by 9 voters

With nine voters, our threshold is $\text{Maj} = \lfloor \frac{9}{2} \rfloor + 1 = 5$. We compute the votes each alternative obtains by counting the number of first votes each alternative obtains in 2.2. No one obtains at least 5 votes and since c had

	a	b	c
$S^{(1)}(x)$	3	4	2

Table 2.2: Vote Count for SVT

the least votes, it is removed from $\mathcal{A}$, hence $\mathcal{A}^{(2)} = \{a, b\}$. The orderings are re-computed and the process repeated in Figure 2.1.

	$\succ^1$	$\succ^2$	$\succ^3$	$\dots$	$\succ^6$	$\succ^7$	$\succ^8$	$\succ^9$
1 st	a	a	a	$\dots$	b	b	a	a
2 nd	b	b	b	$\dots$	a	a	b	b

	a	b	c
$S^{(2)}(x)$	5	4	$\times$

Figure 2.1: Reduced Alternatives Orderings & Vote Count for SVT Round 2

Alternative a meets the threshold and is therefore the winner.

A system originally devised with the goal of creating a representative committee for a small literary association has, in the last two centuries, found its place and use. Since the early 1920s both Ireland and Malta have used Single Transferable Vote for the election of their national legislatures. Beyond them, Australia uses STV for its Senate. In its single-winner formulation, Instant Runoff is used to choose the head of state in Sri Lanka and the national parliament of Papua New Guinea. The states of Alaska and Maine, alongside 39 American cities, also implement Instant Runoff, Tideman (1995); FairVote (2026).

The longevity of the Single Transferable Vote in Ireland and Malta suggests the system is politically durable in ways many alternatives are not; however, it is not without its own pitfalls.

Amongst these failings we find non-monotonicity and the violation of Independence of Irrelevant Alternatives. While these terms are formally defined in Chapter 3, we introduce their intuition below.

Definition 2.4 (Monotonicity). A social aggregation function F is said to be monotone if raising an alternative a in any individual voter’s ranking never lowers a in the societal outcome. Essentially, additional support for an alternative shouldn’t “hurt” its outcome.

Remark 2.5. Monotonicity is a fairly intuitive condition — it requires that a voting mechanism reward support instead of punishing it. In STV a candidate who gains additional first-preference votes may move from winning to losing.

Example 2.6 (Non-monotonicity of STV). Consider an election with three alternatives $\mathcal{A} = \{L, M, R\}$, representing left-wing, moderate, and right-wing respectively, suppose $|V| = 100$, which implies that the winning threshold is 51.

Suppose that individuals who rank L (left) first, always rank the next closest, M (moderate), second. Intuitively, individuals who rank M (moderate) first, always split their second choice equally between L and R . Consider the two profiles outlined in Table 2.3.

Election Profile 1. The first-preference tallies are $L = 31$, $M = 30$, $R = 39$. No alternative meets the threshold, so the lowest-ranked candidate, M , is eliminated. Its 30 votes split equally between L and R , their supporters’ next preferences. This gives $L = 46$ and $R = 54$. **Alternative R is elected.**

	Votes			Runoff Votes			Winner
	L	M	R	L	M	R	
Profile 1	31	30	39	46	—	54	R
Profile 2	25	30	45	—	55	45	M

Table 2.3: Two Profiles for an Election between Alternatives L , M and R

Election Profile 2. Now suppose R gains 6 additional first-preference supporters, defecting from L . The tallies become $L = 25$, $M = 30$, $R = 45$. Again no alternative meets the threshold, and now L has the lowest count and is eliminated. Since left-wing voters prefer the moderate candidate, all 25 of L 's votes transfer to M , giving $M = 55$ and $R = 45$. **Alternative M is elected.**

Despite R gaining support from Profile 1 to Profile 2, R moves from winning to losing. This is a clear violation of monotonicity.

Remark 2.7. The two electoral profiles differ only in that 6 voters have moved from candidate L to candidate R , meaning that R has gained support in the second profile. The outcomes, however, completely change. R won comfortably in the first profile, but by gaining additional supporters from L , it actually loses in the second profile. This is a clear violation of monotonicity.

Definition 2.8 (Independence of Irrelevant Alternatives). A social welfare function F satisfies Independence of Irrelevant Alternatives (IIA) if the societal ranking of any two alternatives a and b depends only on how individual voters rank a against b , and not on where any other alternative appears in their rankings.

Remark 2.9. We encountered IIA informally in Section 1.2.2, where the presence of Ralph Nader altered the outcome between Bush and Gore despite the relative preferences between those two candidates remaining unchanged. STV inherits this same pathology through its transfer mechanism — the elimination of an alternative directly impacts the count of votes that other alternatives have, altering their relative societal ranking.

2.2 Further Attempts at Aggregation

2.2.1 Charles Dodgson’s Voting Rule

While having developed the voting system discussed next, Charles Dodgson is most widely recognised by his pseudonym, Lewis Carroll — the author of *Alice in Wonderland*. Behind the fantastical landscape of Wonderland, the writer, who was a mathematics tutor at Christ Church, Oxford, became interested in methods of voting when getting involved with college appointments. Critical of standard methods of voting already exposed in this body of work, he looked at the weaknesses of the existing mechanisms and proposed new mechanisms to aggregate preferences.

Compelled by Llull’s mechanism and Condorcet’s work, he built on pairwise comparison, as he agreed that comparison was the most logically compelling way to determine the best candidates. Starting from a “condorcet winner” paradigm, he added to this mechanism by asking what should happen when this mechanism led to a contradictory outcome — when the preference cycle that Condorcet had identified rendered pairwise comparison inconclusive. Instead of forgetting about Condorcet’s criterion, he pondered: *who is closest to deserving to win?*

Definition 2.10 (Dodgson Score). Let $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$ be a finite set of alternatives paired with a finite set of voters, $V = [n]$, each handing in a strict linear ordering $>^i$ of all alternatives.

For any alternative $x \in \mathcal{A}$, the Dodgson score $\text{Dod}(x)$ is the minimum number of adjacent swaps (exchanges of two neighbouring alternatives in a single voter’s ranking) required across all ballots to make x a Condorcet winner.

The *Dodgson Winner* is the alternative with the lowest score:

$$\text{Winner} = \arg \min_{x \in \mathcal{A}} \text{Dod}(x)$$

If a Condorcet winner already exists, their score is 0 and they are elected immediately.

Elegantly, Charles Dodgson comes up with a heuristic that in many scenarios can result in a clear winner. There are instances where this formulation fails as would be the case with the profile we explored in Example 1.11, where

we have a three-way cyclic tie and the Dodgson score for each of the alternatives is 1 reflecting the perfect symmetry of the cycle. In said scenario, a tiebreak rule would be required to determine a winner.

An appealing attempt at tackling the voting dilemma, we can still find perils in this methodology. Notably, a big problem with this approach is that it is intractable — the determination of how many adjacent swaps are required to make x a Condorcet winner is NP-hard Caragiannis et al. (2009). This result implies that this problem belongs to a class of computational problems that are at least as hard as the hardest problems in NP and are therefore considered computationally intractable because no polynomial time algorithm exists to solve them.

Another problem inherent in this formulation is its non-homogeneous nature.

Definition 2.11 (Homogeneity). A social welfare system F is homogeneous if duplicating the electorate (replacing every voter’s ballot with two identical ballots) does not change the outcome of the election. Formally, if P is a preference profile and kP denotes the profile in which every ballot appears k times, then $F(P) = F(kP)$ for all positive integers k .

Example 2.12 (Dodgson’s Rule Fails Homogeneity, (Brandt, 2009)). Consider an election with four alternatives $\mathcal{A} = \{a, b, c, d\}$ and $|V| = 12$ voters. We represent the voting state of the election by listing the group size above each column, where each column represents a group of voters with identical preferences listed in descending order:

Group Size	2	2	2	2	2	1	1
1 st	d	b	c	d	a	a	d
2 nd	c	c	a	b	b	d	a
3 rd	a	a	b	c	c	b	b
4 th	b	d	d	a	d	c	c

Table 2.4: Preference profile P with 12 voters. **Alternative a is the Dodgson winner with 3 swaps.**

Alternative a : a already beats b and d . Its only deficit is against c , where it trails 8 to 4, a gap of 4 votes. To flip this, a needs to gain at least

Pair	Votes For	Votes Against	Winner
a vs b	8	4	a
a vs c	4	8	c
a vs d	7	5	a
b vs c	8	4	b
b vs d	6	6	tie
c vs d	6	6	tie

Table 2.5: Pairwise comparisons under profile P .

3 additional supporters over c — that is, it needs $2 + \varepsilon$ swaps in the sense that 3 adjacent switches suffice to move a above c in enough ballots. So $\text{Dod}(a) = 3$.

Alternative d : d ties with b and c . Its only deficit is against a , where it trails 5 to 7. To beat a , alternative d needs just 1 additional supporter, requiring $1 + \varepsilon$ swaps, meaning 2 adjacent switches are enough to flip the a vs d contest. But d must also resolve its ties with b and c , each requiring at least ε — a single swap apiece. In total, $\text{Dod}(d) = 4$.

Since $\text{Dod}(a) = 3 < 4 = \text{Dod}(d)$, alternative a is the Dodgson winner under P .

Now consider the tripled profile $3P$, in which every voter appears three times, giving $|V| = 36$:

Group Size	6	6	6	6	6	3	3
1 st	d	b	c	d	a	a	d
2 nd	c	c	a	b	b	d	a
3 rd	a	a	b	c	c	b	b
4 th	b	d	d	a	d	c	c

Table 2.6: Tripled profile $3P$ with 36 voters. **Alternative d is now the Dodgson winner with 6 swaps.**

Every pairwise margin is now tripled:

Pair	Votes For	Votes Against	Winner
a vs b	24	12	a
a vs c	12	24	c
a vs d	21	15	a
b vs c	24	12	b
b vs d	18	18	tie
c vs d	18	18	tie

Table 2.7: Pairwise comparisons under profile $3P$.

The structure of the election is identical, but the margins have all scaled by a factor of three. This is where homogeneity breaks down. The number of swaps required to flip a pairwise contest scales with the margin — but the number of swaps needed to resolve a *tie* does not since a single adjacent switch suffices regardless of the size of the electorate.

Alternative a under $3P$: The deficit against c has tripled from 4 to 12 votes, so a now requires $6 + \varepsilon$ swaps, meaning $\text{Dod}(a) > 6$.

Alternative d under $3P$: The deficit against a has tripled from 2 to 6 votes, requiring $3 + \varepsilon$ swaps. However, the ties against b and c are still ties — each one requires only ε , a single swap, regardless of how large the electorate has grown. In total, $\text{Dod}(d) = 6$.

Since $\text{Dod}(d) < \text{Dod}(a)$, alternative d is now the Dodgson winner under $3P$, displacing a .

Remark 2.13. It’s worthy of note that Dodgson created this rule as more of a tool to argue against the weaknesses of other methods, than to be implemented in real elections.

Other vulnerabilities inherent in this method include non-monotonicity and its failure to adhere to the Smith Set Principle and to Independence of Clones Brandt (2009).

2.2.2 Edward Nanson’s Method

Unlike other figures that we’ve investigated, Edward Nanson was a professor of mathematics at the University of Melbourne and is now best known for his work in the field of Social Choice.

In 1882, he realised that Borda rule had several advantageous properties, notably, that it is monotonic and that it never elects the Condorcet loser Narodytska et al. (2011). He noted, however, that Borda Count does not necessarily always elect the Condorcet winner. His work set out to remedy this by combining STV in its single winner case and Borda Rule.

Definition 2.14 (Nanson’s Method). Let $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$ and $V = [n]$ as before. At each round r , we compute the Borda score, $S(x)^{(r)}$, given in Definition 1.12 for the remaining alternatives $x \in \mathcal{A}^{(r)}$ as defined in 2.1. The average Borda score for the given round is computed:

$$\bar{S}^{(r)} = \frac{1}{|\mathcal{A}^{(r)}|} \sum_{x \in \mathcal{A}^{(r)}} S^{(r)}(x)$$

All alternatives failing to have the average score or more, $S^{(r)} \leq \bar{S}^{(r)}$ are eliminated at once. The Borda scores are then recomputed on the restricted set, $\mathcal{A}^{(r+1)}$, in the following round.

The process is repeated until a sole alternative remains and is declared the winner.

Evidently, the Condorcet criterion is satisfied by construction given that Condorcet winners always have to score higher than the average and therefore cannot get eliminated. Conveniently, Nanson’s rule is hard to manipulate, especially when compared to regular Borda Count which is famously easy to exploit with simple burying tactics. Nanson’s rule provides a higher barrier for entry for strategic voters, and while, with a smaller number of candidates heuristics can sometimes find manipulative strategies, it can be proven that finding a strategic vote is NP-complete, and therefore cannot be done in polynomial time.

Our investigation of voting systems must continue, however, as Nanson’s Rule inherits many of the pathologies identified in the systems upon which it is built. Amongst its shortcomings, monotonicity and independence of irrelevant alternatives are violated.

2.2.3 Duncan Black's View on Voting

While all of the mechanisms described in this Chapter have shared the common ambition of being able to robustly deal with any possible set of voter preferences, Duncan Black, a Scottish Economist, investigated the conditions under which majority rule works satisfactorily.

Definition 2.15 (Single-Peaked Preferences). A set of voter preferences over a set of alternatives arranged along a single dimension are single-peaked, if each voter has a most preferred alternative and their preference decreases monotonically as alternatives move further from said ideal point in either direction.

Under the condition that all voters present single-peaked preferences, Black was able to prove an interesting result.

Theorem 2.16 (Median Voter Theorem, Black 1948). *If all voter have single-peaked preferences over a one dimensional set of alternatives, $\mathcal{A}$, then a Condorcet winner will always exist, and the winner will be the alternative most preferred by the median voter.*

While we will not prove the Median Voter Theorem here, its intuition makes sense. Any alternative to the left of the median voter will be beaten in a pairwise contest by the median's top preference because the median voter and everyone to their right will form a majority. The argument holds for alternatives to the right of the median voter symmetrically.

Example 2.17. Consider an election with $\mathcal{A} = \{a, b, c, d, e\}$, arranged along a left-to-right ideological spectrum, and three voters whose preferences are illustrated in Figure 2.2. Each voter's preference ranking decreases monotonically away from their ideal candidate, confirming that they are all single-peaked. The median-voter is voter 2 (in blue), whose ideal alternative is b .

We confirm that indeed b is the Condorcet winner:

- b vs a : Voters 2 and 3 prefer b (2 : 1). b wins.
- b vs c : Voters 1 and 2 prefer b (2 : 1). b wins.
- b vs d : All three voters prefer b (3 : 0). b wins.
- b vs e : All three voters prefer b (3 : 0). b wins.

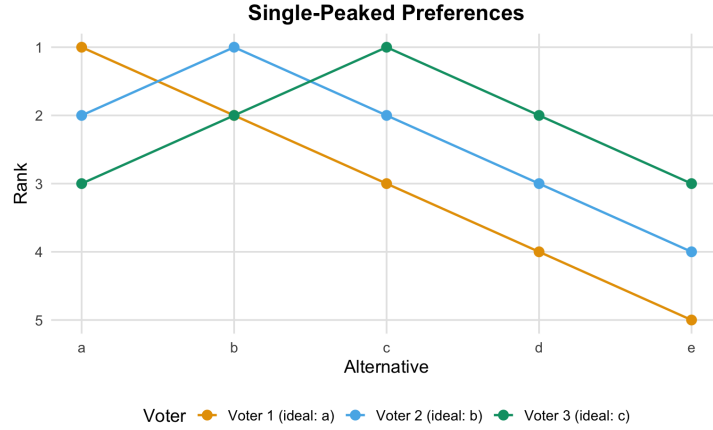


Figure 2.2: Plot of Single-Peaked Preferences. Each line denotes a different voter. The alternative in question is encoded in the x -position and its ranking is encoded in the y axis.

Alternative b — the highest ranked preference of the median voter — defeats every other alternative in pairwise comparison, as suggested by Theorem 2.16.

This work is the first in this entire historical survey that suggests that under certain restricted conditions, we might be able to find genuine compromise. Median Voter Theorem’s assumptions find empirical support in the work of Poole and Rosenthal (1985), who demonstrated that 83% of congressional voting patterns could be predicted using a single ideological dimension: placement along the liberal-conservative spectrum Everson et al. (2016).

Single-peakedness, however, especially in the 21st century, may be too strong of an assumption to make. Political disagreement has become a multi-dimensional phenomenon, scoping economic and social policies which may at times diverge. Similarly, the existence of coalitions may break down voter behaviour, breaking down the guarantees of Black’s theorem and allowing for the resurgence of Condorcet cycles. Black’s theorem, is therefore not a complete solution to the problem of social preference aggregation, but rather a unique characterisation of when these questions behave well.

Black’s identifies a restricted domain under which Arrow’s Impossibility theorem may not apply, but it fails to characterise all possible elections.

The systems explored in this chapter — Single Transferable Vote, Dodgson’s method, Nanson’s method, and Black’s median voter result — represent

some of the most sophisticated attempts at resolving the issues inherent in mechanisms proposed centuries earlier. Each of them made certain strides forwards: STV addressed wasted votes and minority representation; Dodgson restored the Condorcet criterion even in the presence of cycles; Nanson combined Borda's practicality with Condorcet's logical foundation; Black identified the precise condition under which majority rule succeeds. Each fix, presented a new concern: non-homogeneity, monotonicity, computational intractability, or the fragility of its own assumptions.

Even then, the mechanisms surveyed here represent a small fraction of the proposed solutions. Amongst many others, we could have looked into: the Kemeny-Young method, Copeland's method, or Approval voting, which have each attracted scholarly attention. However, each mechanism presented forward has carried its own catalogue of failings. The pattern is not a coincidence of poor design. It is the signature of a deeper mathematical obstruction, one that no amount of clever computation design can work around.

Chapter 3

Democracy's Mathematical Limits

As conveyed by Chapter 1 and Chapter 2, designing a fair voting system is no trivial pursuit. Historical attempts to formulate equitable mechanisms have repeatedly fallen short, plagued by recurring failures; a system that puts out one fire seems invariably threatened by another. We now arrive at the mathematical explanation for why it is inevitably complicated to devise a solution to such a seemingly simple question.

In 1951, Kenneth Arrow proved that when considering two or more alternatives, it is impossible to construct a voting function that simultaneously satisfies a small set of mild fairness axioms Arrow (1950, 1951). This chapter presents a full proof of Arrow's Theorem. In section 3.1 we formalise our problem, introducing the statement of Arrow's results. In section 3.3, we introduce decisive sets, revealing the hidden structures that any system satisfying Arrow's axioms must follow. In Sections 3.4 and 3.5, we show both how power spreads and how it ultimately collapses into a reduced set of individuals. Finally, in 3.6, we complete our argument by discussing the broader implications of the theorem.

3.1 Definitions and Goals

Definition 3.1 (Alternatives). $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$ is a finite set of alternatives to be ranked with $|\mathcal{A}| \geq 3$.

Definition 3.2 (Voters). $V = \{1, 2, \dots, n\}$ is a finite set of voters, $n \geq 2$.

Definition 3.3 (Orderings). Let $\mathcal{B}_{\mathcal{A}}$ denote the set of all possible strict linear orders (complete, transitive) on $\mathcal{A}$. Each element of $\mathcal{B}_{\mathcal{A}}$ represents a possible complete preference ranking a voter may hold.

Definition 3.4 (Preference Profile). A preference profile is a n -tuple representing a set of ballots, $P = (>^1, >^2, \dots, >^n)$ where $>^i \in \mathcal{B}_{\mathcal{A}}$ denotes the preference ordering of voter $i \in V$.

Example 3.5. Suppose $\mathcal{A} = \{a_1, a_2, a_3\}$ and voter 47 prefers a_2 to a_1 to a_3 . We write their ballot as $>^{47}$, which in full would be $a_2 >^{47} a_1 >^{47} a_3$, and note that $>^{47} \in \mathcal{B}$. A preference profile, is a collection of n individual orderings.

Definition 3.6 (Social Welfare Function). A social welfare function is a mapping

$$F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}, \quad (>^1, >^2, \dots, >^n) \mapsto \gg$$

that aggregates individual preferences into a single societal preference order. For a profile $P = (>^1, >^2, \dots, >^n)$ we write $F(>^1, >^2, \dots, >^n) = \gg^P$ for the preference order generated by F on P . For two distinct alternatives, $x, y \in \mathcal{A}$ we write $x \gg^P y$ to indicate that the societal ranking prefers x to y under P . When F and P are fixed, we suppress superscripts and limit ourselves to writing $\gg$ for the societal ranking and $(>^1, >^2, \dots, >^n)$ to denote the set of all voter ballots.

It's worth noting that we require our function F to return a complete and transitive ranking. This is a requirement that the Condorcet's paradox (explored in Section 1.4.1) shows majority rule can fail to satisfy.

Remark 3.7. Some formulations of this question allow the function F to return ties, with individual preferences not necessarily adhering to strict orderings. Arrow's original formulation however, used strict ordering. The impossibility result holds under either convention – working with strict orders simplifies the argument.

3.2 Reasonable Requirements & Axioms

When designing social welfare functions, one may hope to come up with systems meeting a range of appealing and seemingly reasonable conditions — properties that align with our intuitions of what “fair” electoral outcomes

should look like. Some of these properties are motivated and developed in Chapters 1 and 2.

We may hope that our mechanism guarantees anonymity, treating all voters symmetrically and ensuring that no individual has built-in privilege. Ensuring anonymity would imply that switching two ballot's owners (but maintaining the rankings the same) would not alter the result of the function. Similarly, one may hope that alternatives are treated equally, our rules not favouring particular options by name. Alternative characteristics we may want to see our functions satisfy are strategy-proofness, ensuring voters can't obtain a better outcome (from their point of view) by misreporting their preferences, and therefore trying to game the system; or robustness, ensuring that minor adjustments in individual rankings do not produce dramatic swings in social outcomes.

What makes the problem of social choice so surprising is that even when we narrow down our ambitions to a small set of highly intuitive and minimal conditions — namely, Unanimity, Independence of Irrelevant Alternatives and non-dictatorship — we discover that they cannot all be satisfied simultaneously. We explain our axioms and conditions next.

3.2.1 Axiom 1: Unanimity

Definition 3.8 (Unanimity). A social choice function F satisfies *unanimity* if whenever all voters agree on the preference between two alternatives, society agrees with them. That is, if every voter ranks alternative a above b , then society ranks a above b ,

$$(\text{for all } i \in V, a \succ^i b) \implies a \succ b$$

Our function F must respect unanimous agreement. Equivalently, if we let $\Delta : \mathcal{B}_A \rightarrow \mathcal{B}_A^n$ be the diagonal mapping sending a preference order to a constant profile P (a set of ballots where everyone holds the same exact ranking), then unanimity requires the following diagram to commute:

$$\begin{array}{ccc} \mathcal{B}_A & \xrightarrow{\Delta} & \mathcal{B}_A^n \\ & \searrow \text{id} & \downarrow F \\ & & \mathcal{B}_A \end{array}$$

That is, $F \circ \Delta = \text{id}$, our function, when applied to a unanimous profile, simply returns the unanimous ordering.

This condition ensures that our social choice mechanism respects consensus, aligning with unanimous decisions. If everyone prefers one alternative to another, our function should definitely echo the same opinion.

3.2.2 Axiom 2: Independence of Irrelevant Alternatives

Definition 3.9 (Independence of Irrelevant Alternatives, IIA). A social choice function satisfies IIA if the social ranking between any two alternatives depends solely on individual preferences between those two alternatives. In other words, the social ranking of any pair $\{a, b\}$ depends solely on how voters rank a versus b , independently of how they ranked other alternatives.

Let $P^{a>b} = \{i \in V \mid a \succ^i b \text{ in profile } P\}$ denote the coalition of voters who prefer a to b under P . Formally, we say that fixed F satisfies IIA if for any two profiles $P, Q \in \mathcal{B}_{\mathcal{A}}^n$:

$$\text{If } P^{a>b} = Q^{a>b}, \text{ then } a \gg^P b \iff a \gg^Q b$$

Independence of Irrelevant Alternatives is a powerful condition as it states that the social ordering on any pair is determined entirely by which group of voters prefer a over b , regardless of where any of the other alternatives appear in their rankings.

Proposition 3.10 (IIA via Restriction Maps). *For any $X \subseteq \mathcal{A}$, define $r_X^A : \mathcal{B}_{\mathcal{A}} \rightarrow \mathcal{B}_X$ as the map forgetting the relative positions of alternatives outside of X . A social welfare function $F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$ satisfies IIA if and only if for every pair of distinct alternatives $\{a, b\} \subseteq \mathcal{A}$, there exists a function $F' : \mathcal{B}_{\{a,b\}}^n \rightarrow \mathcal{B}_{\{a,b\}}$ such that the following diagram commutes:*

$$\begin{array}{ccc} \mathcal{B}_{\mathcal{A}}^n & \xrightarrow{F} & \mathcal{B}_{\mathcal{A}} \\ \downarrow (r_X^A)^n & & \downarrow r_X^A \\ \mathcal{B}^n & \xrightarrow{F'} & \mathcal{B}_X \end{array}$$

That is, restricting the societal output to X yields the same results as first restricting each voter's ballot to X and then applying the local rule F' . Equivalently, F commutes with restrictions to subsets of alternatives.

Example 3.11. Consider three voters ranking five individuals, $\mathcal{A} = \{a, b, c\}$. One possible set of rankings could be profile P :

	$>^1$	$>^2$	$>^3$	$>^4$	$>^5$
1 st	a	a	b	b	c
2 nd	b	b	c	c	a
3 rd	c	c	a	a	b

Using Borda Count to come up with a societal ranking:

- Given that a receives two 1st choice votes; one 2nd choice vote and two 3rd choice ranking, a gets $(2 \times 2) + (1 \times 1) + (2 \times 0) = 5$ points.
- Similarly, b receives $(2 \times 2) + (2 \times 1) + (1 \times 0) = \mathbf{6}$ points.
- Then, c receives $(1 \times 2) + (2 \times 1) + (2 \times 0) = 4$ points.

Our Borda Count would dictate:

$$b \gg a \gg c$$

That is, b is preferred to a which is preferred to c .

Now consider restricting the set of alternatives to $\mathcal{A} = \{a, b\}$, and constructed profile Q to maintain the relative orderings of all other alternatives, we would get:

	$>^1$	$>^2$	$>^3$	$>^4$	$>^5$
1 st	a	a	b	b	a
2 nd	b	b	a	a	b

Our Borda Count mechanism would now compute:

- a receives $(3 \times 1) + (2 \times 0) = \mathbf{3}$ points.
- b receives $(2 \times 1) + (3 \times 0) = 2$ points.

Notice that now, a receives the most points, as such our Borda Count results in:

$$a \gg b$$

Note that every voter ranked a and b the same way in both profiles: $P^{a>b} = Q^{a>b} = \{>^1, >^2, >^5\}$, yet the social ranking of a vs b changed relative to c . In this case, Borda Count is demonstrated to violate IIA.

Definition 3.12 (Fair System). We determine that a social welfare function $F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$ is *fair* if it satisfies both Unanimity and Independence of Irrelevant Alternatives.

Theorem 3.13 (Arrow's Impossibility Theorem). *Consider an election where $|\mathcal{A}| \geq 3$ and $|V| \geq 2$. Every fair social welfare function $F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$ is a dictatorship. That is for any F satisfying Unanimity and IIA, there exists a voter $d \in V$ such that F 's rule, $\gg$ matches $>_d$ for any profile P .*

Remark 3.14 (Universal Domain). We implicitly assume that F is defined on the full domain $\mathcal{B}_{\mathcal{A}}^n$. In other words we assume that F must be able to accept any combination of individual preference orders as input. This condition, known as *Universal Domain*, is a standing assumption throughout this work.

Remark 3.15 (Monotonicity). We say that a social welfare function F is *monotone* if raising an alternative a in any voter's ranking does not lower a in the societal output. It can be shown that any F satisfying Unanimity and IIA is necessarily monotone, so monotonicity could equivalently replace one of these conditions in the hypotheses above.

The rest of this chapter is devoted to proving Theorem 3.13.

3.3 What is Decisiveness?

Arrow's theorem argues that certain coalitions of voters have the power to determine the social ranking of a pair. Our goal will be to demonstrate how IIA ensures that this power depends only on said coalitions and not on the profiles under evaluation. To this end, we define decisive sets next and prove their properties in 3.3.1.

Definition 3.16 (*Weakly Decisive Set*). A coalition $S \subseteq V$ is weakly decisive for a over b (written $S \in \mathcal{D}_{a>b}$) if, whenever every voter in S prefers a to b and every voter outside prefers b to a , society ranks a above b . That is:

$$S \in \mathcal{D}_{a>b} \iff [\text{for all } i \in S, a >^i b \text{ and for all } j \notin S, b >^j a] \implies a \gg b$$

Lemma 3.17 (Characterisation of Weak Decisiveness via IIA). *Let F satisfy IIA. If there exists a profile P with $P^{a>b} = S$ such that $a \gg^P b$, then $S \in \mathcal{D}_{a>b}$.*

Proof. Let Q be any profile such that $Q^{a>b} = S$, meaning every voter in S prefers a to b and every voter outside S prefers b to a . Since $P^{a>b} = Q^{a>b} = S$, IIA guarantees that $a \gg^P b \iff a \gg^Q b$. Since $a \gg^P b$ by hypothesis, we conclude $a \gg^Q b$. As Q was an arbitrary profile with $Q^{a>b} = S$, this holds for all such profiles, so $S \in \mathcal{D}_{a>b}$. $\square$

Remark 3.18. One might hope that weak decisiveness is preserved under taking supersets: if S can force $a \gg b$ against all opposition, surely a larger coalition (having more support) $S' \supseteq S$ can too. This would follow from monotonicity of F , but we do not assume this a priori. We therefore introduce a stronger notion of decisiveness that does not rely on the behaviour of voters outside S .

Definition 3.19 (Strongly Decisive). A coalition $S \subseteq V$ is strongly decisive for a over b (written $S \in \mathcal{D}^*_{a>b}$) if, whenever every voter in S prefers a to b (regardless of what voters outside S do), society ranks a over b :

$$S \in \mathcal{D}^*_{a>b} \iff [\text{for all } i \in S, a >^i b] \implies a \gg b$$

It follows that strong decisiveness implies weak decisiveness, so $\mathcal{D}^*_{a>b} \subseteq \mathcal{D}_{a>b}$. What is most surprising about Arrow's theorem is that under our conditions, these two coincide, and that we eventually show that they are independent of the pair (a, b) .

Remark 3.20 (IIA helps us define Decisiveness). By satisfying IIA, we ensure that any function's societal ranking of a above b depends only on the set of people holding that preference, $P^{a>b}$. To test whether $S \in \mathcal{D}_{a>b}$, it suffices to exhibit any single profile P with $P^{a>b} = S$ and verify whether the output $a \gg b$. If it holds for one such profile, it must hold for all.

3.3.1 Properties of Decisive Sets

Proposition 3.21 (Monotonicity & Closure Under Supersets). *If $S \in \mathcal{D}_{a>b}$ and $S \subseteq T$, then $T \in \mathcal{D}_{a>b}$.*

Proof. This is implicit in the proof of Corollary 3.27. $\square$

Proposition 3.22 (Complement Property). *For any $S \subseteq V$:*

$$S \in \mathcal{D}_{a>b} \iff S^c = V \setminus S \notin \mathcal{D}_{b>a}$$

Proof. Consider a profile P in which $P^{a>b} = S$, which implies $S^c = P^{b>a}$. Since our function must give a total ordering, exactly one of $a \gg b$ or $b \gg a$ holds. By IIA, which one holds depends only on S . If $a \gg b$, then $S \in \mathcal{D}_{a>b}$ and $S^c \notin \mathcal{D}_{b>a}$. $\square$

Proposition 3.23 (Unanimity Implies V is Decisive). *$V \in \mathcal{D}_{a>b}$ for every pair $a \neq b$, and $\emptyset \notin \mathcal{D}_{a>b}$ for any pair.*

Proof. The first statement follows directly from the axiom of unanimity. The second axiom is implied by the complement property: $V \in \mathcal{D}_{a>b}$ implies $V^c = \emptyset \notin \mathcal{D}_{a>b}$. Since this is true for any a, b , it follows that $\emptyset$ is not decisive for any pair. $\square$

3.4 Decisive for One Pair, Implies Decisive for All Pairs

Having defined our axioms and our decisiveness terms, along with their properties, we start observing the moving cogs of this proof. The following lemma shows how decisiveness “spreads” via the transitivity of our social ordering mechanisms.

3.4.1 Contagion

Lemma 3.24 (Contagion). *If $A \in \mathcal{D}_{a>b}$ and $B \in \mathcal{D}_{b>c}$, then $A \cap B \in \mathcal{D}_{a>c}$, where a, b, c are distinct alternatives.*

Proof. If we assume that $A \in \mathcal{D}_{a>b}$, it follows that $a \gg b$. Similarly, if we assume that $B \in \mathcal{D}_{b>c}$, then we know $b \gg c$. It follows by transitivity,

$$a \gg b \gg c \implies a \gg c$$

The individuals that held this exact preference must necessarily be those in A and B . Therefore, since $a \gg c$, and the individuals that held this preference are exactly those in $A \cap B$, it follows that $A \cap B \in \mathcal{D}_{a>c}$. This exact relationship can be observed in the venn diagram in Figure 3.1. $\square$

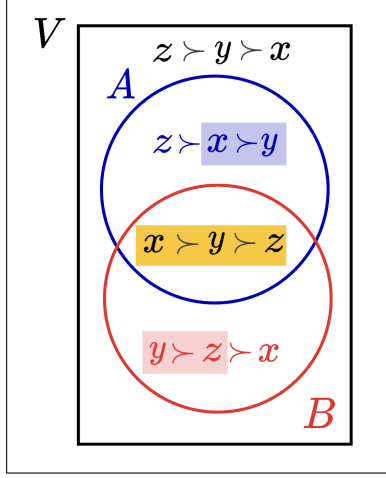


Figure 3.1: Voters in A (blue) prefer $a > b$, and voters in B (red) prefer $b > c$. Voters in $A \cap B$ (gold) hold both preferences, ranking $a > b > c$. By transitivity, $A \cap B$ is the coalition that prefers $a > c$ illustrating that $A \cap B \in \mathcal{D}_{a>c}$.

Example 3.25. Consider $V = \{>^1, >^2, >^3, >^4, >^5\}$, $\mathcal{A} = \{a, b, c\}$. Suppose $A = \{>^1, >^2, >^3\}$ is decisive for $a > b$ and $B = \{>^2, >^3, >^4\}$ is decisive for $b > c$. The ballots are displayed in Table 3.1:

	$>^1$	$>^2$	$>^3$	$>^4$	$>^5$
1 st	c	a	a	b	c
2 nd	a	b	b	c	b
3 rd	b	c	c	a	a

Table 3.1: Ballots for Example 3.25

Voter 1 is in $A \setminus B$, voters 2 and 3 are in $A \cap B$, voter 4 in $B \setminus A$, and voter 5 in $(A \cup B)^c$. By decisiveness of A and B on their respective pairs of alternatives and using transitivity, we conclude that $a \gg c$ with only $A \cap B = \{>^2, >^3\}$ supporting this.

3.4.2 Decisive for One, Decisive for All

There is a result that naturally follows from our contagion lemma: a coalition that is weakly decisive for any one pair, is actually strongly decisive for every pair.

Lemma 3.26 (Weakly Decisive for One, Strongly Decisive for All). *If $S \in \mathcal{D}_{a>b}$ for some pair a, b , then $S \in \mathcal{D}_{x>y}$ for all pairs x, y of distinct alternatives.*

Proof. Let S be weakly decisive for a over b , $S \in \mathcal{D}_{a>b}$, and $c \in \mathcal{A} \setminus \{a, b\}$. By Unanimity, $V \in \mathcal{D}_{b>c}$, so, by Contagion, $S = S \cap V \in \mathcal{D}_{a>c}$. Similarly, $V \in \mathcal{D}_{c>a}$, so by Contagion, $V \cap S = S \in \mathcal{D}_{c>b}$. Following this logic, we get:

$$\begin{aligned} S \in \mathcal{D}_{a>b} &\iff S \in \mathcal{D}_{a>c} \\ &\iff S \in \mathcal{D}_{c>b} \\ &\iff S \in \mathcal{D}_{b>c} \\ &\iff S \in \mathcal{D}_{b>a} \\ &\iff S \in \mathcal{D}_{c>a} \end{aligned}$$

Since $c \in \mathcal{A} \setminus \{a, b\}$ was arbitrary, this establishes weak decisiveness for all pairs involving at least one of a or b .

We must still show that decisiveness holds for pairs $\{c, d\}$ where neither c nor d are equal to a or b . Since $c \neq a$, the pair $\{c, a\}$ involves a , so by the above $S \in \mathcal{D}_{c>a}$. Since $d \neq a$ the pair $\{a, d\}$ also involves a , therefore, similarly, $S \in \mathcal{D}_{a>d}$. Applying Contagion Lemma 3.24, to $S \in \mathcal{D}_{c>a}$ and $S \in \mathcal{D}_{a>d}$, we obtain $S \in \mathcal{D}_{c>d}$.

Since c and d were arbitrary distinct alternatives in $\mathcal{A} \setminus \{a, b\}$, we conclude that S is weakly decisive for every pair of distinct alternatives in $\mathcal{A}$. $\square$

Corollary 3.27 (Weak Decisiveness for One, implies Strong Decisiveness for All). *If a set S is weakly decisive for any pair of distinct alternatives, then S is strongly decisive for every pair of distinct alternatives.*

Proof. By Lemma 3.17, the hypothesis that $S \in \mathcal{D}_{a>b}$ means there exists a profile P with $P^{a>b} = S$ under which $a \gg^P b$. By IIA, this extends to all profiles with the same coalition, so S forces $a \gg b$ regardless of what voters outside S do. Combined with Lemma 3.26, S is weakly decisive for every pair, and since it overcomes all opposition in each case, strong decisiveness follows. $\square$

Lemma 3.26 together with Corollary 3.27 justify calling a coalition decisive (without specifying any pair) once it is known that it is weakly decisive for any single pair. We therefore simplify our notation to say

$$\mathcal{D} = \mathcal{D}_{a>b} \text{ for any pair } a \neq b$$

3.5 Power Collapses into Individuals

We have shown that once a coalition is decisive for a pair of alternatives, it is decisive for the social ranking of all of our options. We now show that the decisive structure reduces to a single voter.

Definition 3.28 (Minimal Decisive Set). Since $\mathcal{D}$ is non-empty (by Proposition 3.23) and V is finite, there exists a decisive set of minimal cardinality. We fix one and call it M :

$$M = \arg \min_{S \in \mathcal{D}} |S|$$

By Proposition 3.23, $|M| \geq 1$ since $\emptyset \notin \mathcal{D}$.

Theorem 3.29 (A Dictator). $|M| = 1$. *There exists a single voter $d \in V$ such that $\{d\}$ is decisive.*

Proof. Suppose by way of contradiction, that $|M| \geq 2$. It follows that we can partition M into two non-empty disjoint subsets, R and S . $T = V \setminus M$ represents all voters outside of our minimal decisive set. R, S are strictly smaller than M , therefore by the minimality of M , neither R or S are decisive.

Consider three distinct alternatives $a, b, c \in \mathcal{A}$, and consider the following profile presented in Figure 3.2.

Since M is our decisive set, and every voter in M (both R and S) prefers $a \succ^M b$, it follows that society must rank $a \succ b$.

We then consider the societal ranking between a and c . The only voters who prefer a to c belong to R . It follows, that if our societal ranking resulted in $a \succ c$, this would imply that R is a weakly decisive set for a over c . By Corollary 3.27, R would be decisive for all pairs. However, since $R \subsetneq M$, we have $|R| < |M|$. This statement would directly contradict our premise that M is the decisive set of minimal cardinality. It follows that our output cannot choose $a \succ c$. Given that F must give a strict complete ordering, the

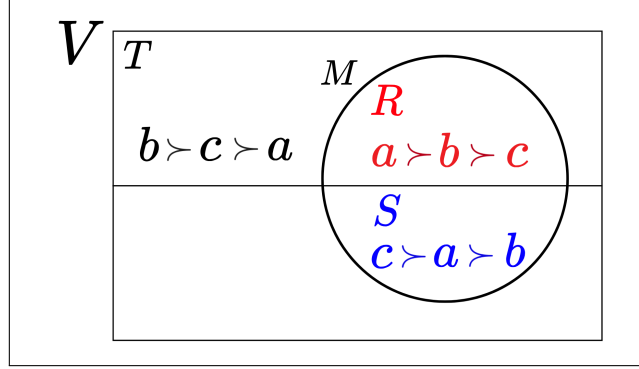


Figure 3.2: Preferences of people in R , S and T

only other output would be for our social ranking to prefer c to a . Therefore, $c \gg a$.

We now know that $c \gg a$ and $a \gg b$. Therefore, by transitivity, we would have $c \gg a \gg b$, which would imply $c \gg b$.

Taking a look at individual preferences for c versus b , we realise that the only voters that preferred c to b are those in S . Given that the societal outcome is $c \gg b$, it follows that S is weakly decisive of c over b . Again, by 3.27, S becomes a decisive set over all alternatives.

However S is, like R , a strict subset of M , implying that $|S| \leq |M|$. This contradicts the minimality of M .

Since assuming $|M| \geq 2$ inevitably leads us to a contradiction regarding the minimality of M , our assumption must be false. It follows that $|M| < 2$. We know that any decisive set must be non-empty by Proposition 3.23, therefore $0 < |M| \implies |M| = 1$. The minimal decisive set consists of exactly one voter, d . Because this single voter is decisive for all pairs, the social ordering $\gg$ will always match $\succ^d$, fulfilling the definition of a dictator. $\square$

3.6 Dictatorship is Inevitable

Proof (Arrow's Impossibility Theorem, 3.13). Let $F : \mathcal{B}^n \rightarrow \mathcal{B}$ satisfy Unanimity 3.8 and Independence of Irrelevant Alternatives 3.9. By Proposition 3.23, given that we know that V is decisive, we know that $\mathcal{D}$ is non-empty. By Theorem 3.29, there exists $d \in V$ such that $\{d\} \in \mathcal{D}$. By Lemma 3.26, we know that $\{d\}$ is strongly decisive for every pair, which implies that for any

$P \in \mathcal{B}^n$, and any $a \neq b$,

$$a \succ^d b \implies a \gg b$$

This means that our function copies d 's strict ranking for any input. It follows that d is a dictator. $\square$

In this chapter, we gave a complete proof of Arrow's Impossibility Theorem. We first formalised the problem of social choice by introducing the axioms of Unanimity and Independence of Irrelevant Alternatives, and then demonstrated that any fair social welfare function satisfying both must be a dictatorship. The proof proceeded by first introducing decisive sets which are given to us by Unanimity. We then implemented IIA to show how decisiveness depends only on coalitions of individuals and not different profiles. This allowed us to demonstrate via contagion that decisiveness spreads across all pairs. Finally, we were able to show that minimality of these decisive sets collapses into single voters.

What is surprising is not the fact that no fair system exists, but that the only logically consistent outcome is the concentration of all power in a single individual. Through this surprising result, Arrow allows us to understand why all the mechanisms explored in Chapters 1 and 2 were doomed to fail.

In the upcoming chapter we revisit this theorem through the lens of ultrafilters, offering a structurally cleaner perspective that makes plain why this collapse to dictatorship is not a coincidence, but an inevitability.

Chapter 4

A Topological Approach: Ultrafilters and Dictators

We now present an alternative proof of Arrow's Impossibility Theorem (Theorem 3.13) using the notion of *ultrafilters*, following the approach of Goldbring (2022). Rather than tracing how power spreads through decisive coalitions, this approach reveals why a dictator must exist as an immediate consequence of a classical fact: every ultrafilter on a finite set is principal. Along the way, we also identify the unique scenario under which a non-dictatorial solution becomes possible.

4.1 Filters

Here we introduce the necessary terminology and concepts required to develop this argument, under the same electoral context established in Chapter 3. Throughout, $V = \{1, \dots, n\}$ denotes our fixed finite set of voters and $\mathcal{A}$ denotes the set of alternatives with $|\mathcal{A}| \geq 3$.

Definition 4.1 (Filter). A *filter* on a set of voters V , is a non-empty collection of sets, $\mathcal{F} \subseteq \mathcal{P}(V)$ (where $\mathcal{P}$ denotes the power set) that satisfies the following conditions:

- (1) $\emptyset \notin \mathcal{F}$, and, $V \in \mathcal{F}$;
- (2) If $S, T \in \mathcal{F}$, then $S \cap T \in \mathcal{F}$;
- (3) If $S \in \mathcal{F}$ and $S \subseteq T$, then $T \in \mathcal{F}$;

We think of sets belonging to $\mathcal{F}$ as “large” coalitions of voters. The first condition reflects that V is the largest possible coalition, hence being contained in $\mathcal{F}$, whereas the empty set is quite the opposite. The third condition — upwards closure — implies that any coalition containing a large set is itself large and therefore in $\mathcal{F}$. The second condition, that of intersection, is perhaps the most counterintuitive. It asserts that if coalition S is large enough to be decisive, and coalition T is large enough to be decisive, then their overlap $S \cap T$ must be large enough to be decisive as well. This implies that two large enough coalitions cannot be entirely disjoint — they must share a common core of voters which in itself is large.

Example 4.2. The collection $\mathcal{F} = \{V\}$ is the smallest possible filter on V , containing the full voting population. On the other side of the spectrum, $\mathcal{P}(V) \setminus \{\emptyset\}$ is also a filter.

Example 4.3. (Is a “Majority” a Filter? Suppose $|V| = 5$, so $V = \{1, 2, 3, 4, 5\}$. We define a set that describes a “majority”:

$$\mathcal{F}_{\text{maj}} := \{S \subseteq V : |S| \geq 3\}$$

We notice when trying to verify the filter axioms:

- (1) $V \in \mathcal{F}_{\text{maj}}$ since $|V| = 5 \geq 3$, and $\emptyset \notin \mathcal{F}_{\text{maj}}$ since $|\emptyset| = 0 < 3$.
- (2) If $S, T \in \mathcal{F}_{\text{maj}}$, then $|S| \geq 3$ and $|T| \geq 3$. Does $S \cap T \in \mathcal{F}_{\text{maj}}$? Not always — consider $S = \{1, 2, 3\}$ and $T = \{3, 4, 5\}$; $S \cap T = \{3\}$, hence $|S \cap T| = 1 \not\geq 3$, hence $S \cap T \notin \mathcal{F}_{\text{maj}}$.

It therefore follows that our definition $S \cap T \in \mathcal{F}_{\text{maj}}$ fails at least one of our axioms and is not a filter. This exemplifies how filters are non-trivial — our notion of majority doesn’t automatically form a filter.

4.2 Ultrafilters

In our voting setting, we will think of a set $S \in \mathcal{F}$ as a *decisive* coalition — one large enough that, if members agree, their collective preference could determine the social outcome. A filter is not yet a complete classification of coalitions in this sense: some coalitions S may be unclassified, with neither S or $V \setminus S$ declared as decisive. An ultrafilter resolves this, acting as a decisive judge that takes an unambiguous stance on every subset of V — every

coalition will either be decisive or not, with no space in between. As we will see, the collection of decisive coalitions of any fair social welfare function will turn out to carry exactly this structure.

Definition 4.4 (Ultrafilter). A filter $\mathcal{U}$ on V is a *ultrafilter*, if for every $S \subseteq V$, either $S \in \mathcal{U}$ or $S^c = V \setminus S \in \mathcal{U}$, but not both.

An ultrafilter is therefore a complete and consistent verdict: every coalition is unambiguously declared large enough or not.

Proposition 4.5 (Maximality Criterion). *A filter $\mathcal{U}$ on V is an ultrafilter if and only if it is maximal: there is no filter $\mathcal{F}$ on V with $\mathcal{U} \subsetneq \mathcal{F}$.*

This is an alternative characterisation or definition for any ultrafilter.

Proof. ($\Rightarrow$) Assume that $\mathcal{U}$ is an ultrafilter. Let $\mathcal{F}$ be a filter such that $\mathcal{U} \subseteq \mathcal{F}$. Take any $S \in \mathcal{F}$. Since $\mathcal{U}$ is an ultrafilter, either $S \in \mathcal{U}$ or $S^c \in \mathcal{U}$. If $S^c \in \mathcal{U}$, then $S \cap S^c = \emptyset \in \mathcal{F}$ contradicting at least one of the conditions of filters. Hence $S \in \mathcal{U}$. Thus $\mathcal{F} \subseteq \mathcal{U}$ and $\mathcal{F} = \mathcal{U}$.

($\Leftarrow$) Suppose $\mathcal{U}$ is maximal but not an ultrafilter. It follows there exists $S \subseteq V$ such that neither S nor S^c belongs to $\mathcal{U}$. Define:

$$\mathcal{F}^S := \{R \subseteq V \mid \exists F \in \mathcal{F} \text{ such that } F \cap S \subseteq R\}$$

We'll argue that $\mathcal{F}^S$ is a filter on V strictly containing $\mathcal{U}$ contradicting maximality. We verify the filter axioms:

- (1) Given that $\mathcal{U}$ is a filter, $V \in \mathcal{U}$. Therefore, letting $F = V$ gives $S \cap V = S \subseteq V$, so $V \in \mathcal{F}^S$.

Consider $\emptyset \in \mathcal{F}^S$, then $S \cap F \subseteq \emptyset$ for some $F \in \mathcal{U}$, meaning $S \cap F = \emptyset$, so $F \subseteq S^c$. Since $F \in \mathcal{U}$ and $\mathcal{U}$ is closed upward, this would give $S^c \in \mathcal{U}$, contradicting our assumption. Hence $\emptyset \notin \mathcal{F}^S$.

- (2) If $T_1, T_2 \in \mathcal{F}^S$, there exists $F_1, F_2 \in \mathcal{U}$ with $S \cap F_1 \subseteq T_1$ and $S \cap F_2 \subseteq T_2$. Since $F_1 \cap F_2 \in \mathcal{U}$ and $S \cap (F_1 \cap F_2) \subseteq T_1 \cap T_2$, we get $T_1 \cap T_2 \in \mathcal{F}^S$.
- (3) If $T \in \mathcal{F}^S$ and $T \subseteq T'$, then $S \cap F \subseteq T \subseteq T'$ for some $F \in \mathcal{U}$, so $T' \in \mathcal{F}^S$.

Hence $\mathcal{F}^S$ is a filter. Moreover $\mathcal{U} \subseteq \mathcal{F}^S$ (given any $F \in \mathcal{U}$, take F itself: $S \cap F \subseteq F$), and $S \in \mathcal{F}^S$ (take $F = V$: $S \cap V = S \subseteq S$) while $S \notin \mathcal{U}$. Therefore, $\mathcal{U} \subsetneq \mathcal{F}^S$, contradicting maximality. $\square$

4.3 Principal Ultrafilters

Definition 4.6 (Principal Ultrafilter). For any voter $d \in V$, principal ultrafilter generated by d is the coalition

$$\mathcal{U}_d := \{S \subseteq V : d \in S\}$$

This declares a coalition “large” if and only if it contains d .

Proposition 4.7. *For any voter $d \in V$, the set $\mathcal{U}_d$ is an ultrafilter on V .*

Proof. We proceed to check that $\mathcal{U}_d$ satisfies all of the filter axioms and the ultrafilter condition.

- (1) Since $d \in V$, we have that $V \in \mathcal{U}_d$. Since $\emptyset$ contains no voters, $d \notin \emptyset$, which implies $\emptyset \notin \mathcal{U}_d$.
- (2) If $S \in \mathcal{U}_d$ and $T \in \mathcal{U}_d$, then $d \in S$ and $d \in T$. It follows that $d \in S \cap T$, so $S \cap T \in \mathcal{U}_d$.
- (3) If $S \in \mathcal{U}_d$, by definition, $d \in S$. If $S \subseteq T$ and $d \in S$, then $d \in T$, which implies $T \in \mathcal{U}_d$.

Ultrafilter Condition. For any $S \subseteq V$, we show exactly one of $S \in \mathcal{U}_d$ or $S^c \in \mathcal{U}_d$ holds.

- ($\Rightarrow$) If $S \in \mathcal{U}_d$, then $d \in S$, which implies $d \notin S^c$, and hence $S^c \notin \mathcal{U}_d$.
- ($\Leftarrow$) If $S \notin \mathcal{U}_d$, then $d \notin S$, which implies $d \in S^c$, and hence $S^c \in \mathcal{U}_d$.

Hence $\mathcal{U}_d$ is an ultrafilter on V . $\square$

Example 4.8. Consider $V = \{1, 2, 47\}$, consider the principal ultrafilter $\mathcal{U}_{47}$ generated by voter 47. The sets belonging to $\mathcal{U}_{47}$ are precisely those containing voter 47:

$$\mathcal{U}_{47} = \{\{47\}, \{1, 47\}, \{2, 47\}, \{1, 2, 47\}\}$$

The excluded sets are those that don't contain the element 47, so $V \setminus \mathcal{U}_{47} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. One can check that for each of these excluded sets, S , the complement belongs to $\mathcal{U}_{47}$.

Lemma 4.9. *If $\{d\} \in \mathcal{U}$ for some $d \in V$, then $\mathcal{U} = \mathcal{U}_d$.*

Proof. Take any $S \subseteq V$. If $d \in S$, then $\{d\} \subseteq S$, and since $\{d\} \in \mathcal{U}$, upward closure gives $S \in \mathcal{U}$. If $d \notin S$, then $d \in S^c$, so $S^c \in \mathcal{U}$, and the ultrafilter condition forces $S \notin \mathcal{U}$. Hence $S \in \mathcal{U} \iff d \in S$, which is precisely $\mathcal{U}_d$. $\square$

The following proposition is the pivotal fact that will force our dictator to exist.

Proposition 4.10. *Every ultrafilter on a finite set V is principal. That is, there exists $d \in V$ such that $\mathcal{U} = \mathcal{U}_d$.*

Proof. Let $\mathcal{U}$ be an ultrafilter on a finite set of voters V . Define

$$D := \bigcap_{S \in \mathcal{U}} S$$

We claim that $D \in \mathcal{U}$. Indeed, since V is finite its power set is finite so $\mathcal{U}$ contains only finitely many sets. As $\mathcal{U}$ must be closed under finite intersections, this finite intersection belongs to $\mathcal{U}$, meaning $D \in \mathcal{U}$. By our first filter axiom $\emptyset \notin \mathcal{U}$, hence $D \neq \emptyset$.

We will argue that D is a singleton. For the sake of contradiction, assume that $|D| \geq 2$, pick any $d \in D$. By the ultrafilter property, we know that either $\{d\} \in \mathcal{U}$ or $\{d\}^c = V \setminus \{d\} \in \mathcal{U}$.

- If $\{d\} \in \mathcal{U}$, since D is the intersection of all elements in $\mathcal{U}$, then it must be $D \subseteq \{d\}$, in that case $|D| = 1$, contradicting our assumption $|D| \geq 2$.
- Alternatively, if $\{d\}^c = V \setminus \{d\} \in \mathcal{U}$, since $D \in \mathcal{U}$ and $\mathcal{U}$ is closed under intersection, then $D \cap V \setminus \{d\} \in \mathcal{U}$. Notice that $D \cap V \setminus \{d\} = D \setminus \{d\}$, which is strictly smaller than D and therefore a proper subset. Repeating this argument, removing one element at a time eventually gives us $\emptyset \in \mathcal{U}$ contradicting our first axiom for the definition of a filter.

It follows that $|D| = 1$, meaning $D = \{d\}$ for some unique $d \in V$, giving $\{d\} \in \mathcal{U}$. By Lemma 4.9, $\mathcal{U} = \mathcal{U}_d$.

We now show that $\mathcal{U} = \mathcal{U}_d$. Recall that filters are closed under supersets. Take any $S \subseteq V$. If $d \in S$, then $\{d\} \subseteq S$, and since $\{d\} \in \mathcal{U}$, upward closure gives us that $S \in \mathcal{U}$. Alternatively, if $d \notin S$, then $d \in S^c$, that would imply $S^c \in \mathcal{U}$ and the ultrafilter property would then force $S \notin \mathcal{U}$.

Hence, $S \in \mathcal{U}$ if and only if $d \in S$, which is the exact definition of $\mathcal{U}_d$. $\square$

4.4 Proving Arrow’s Theorem via Ultrafilters

Having laid down the foundations of Ultrafilters, we can now connect them to social welfare functions. Readers familiar with Chapter 3 may recall that a coalition $S \subseteq V$ was called weakly decisive for a over b if the unanimous preference of S could force the social outcome, even against unanimous opposition from $V \setminus S$. The proof in Chapter 3 proceeded by showing that such power “spreads” and ultimately collapses into a single voter’s preferences.

The argument we are about to present takes a different approach entirely. We consider all decisive coalitions at once as a single mathematical object and ask what structure that collection must have. We will argue that Unanimity and IIA alone force this set to be an ultrafilter — and on a finite set, every ultrafilter is principal.

4.4.1 Ultrafilters for “Fair” Social Welfare Functions

We start by showing that any ultrafilter on V will naturally define a *fair* social welfare function.

Definition 4.11 (Ultrafilter Social Welfare Function). Let $\mathcal{U}$ be an ultrafilter on V . Define $F_{\mathcal{U}} : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$ by specifying, for each profile $P = (>^1, >^2, \dots, >^n)$, the societal preference order $F_{\mathcal{U}}(P) = \gg$ by:

$$a \gg b \iff \{i \in V : a >^i b\} \in \mathcal{U}$$

for any distinct $a, b \in \mathcal{A}$. That is, society prefers a to b if and only if the coalition of voters preferring a to b is “large” according to $\mathcal{U}$.

Theorem 4.12. *For any ultrafilter $\mathcal{U}$ on V , the relation $F_{\mathcal{U}}(P) = \gg$ is a total order on $\mathcal{A}$, and $F_{\mathcal{U}}$ is a **fair** social welfare function — that is, it satisfies Unanimity and Independence of Irrelevant Alternatives.*

Proof. We verify each of the properties.

To ensure that any output is *totally ordered*, we need to check for completeness, antisymmetry, and transitivity.

We know that the output is complete, because, given any distinct $a, b \in \mathcal{A}$, the sets $\{i : a >^i b\}$ and $\{i : b >^i a\}$ partition V , hence by the ultrafilter condition only one belongs to $\mathcal{U}$, this gives exactly one of $a \gg b$ or $b \gg a$. Antisymmetry follows from this result.

Suppose that the societal ordering gives $a \gg b$ and $b \gg c$, it follows, $A := \{i : a >^i b\} \in \mathcal{U}$ and $B := \{i : b >^i c\} \in \mathcal{U}$. Then $A \cap B \in \mathcal{U}$ by closure under intersection, and every voter in $A \cap B$ ranks $a >^i b >^i c$, so $A \cap B \subseteq \{i : a >^i c\}$. By upward closure, $\{i : a >^i c\} \in \mathcal{U}$, giving $a \gg c$. Hence, the relation is a total order on all the alternatives, $\mathcal{A}$.

If every voter prefers a to b , then $\{i : a >^i b\} = V \in \mathcal{U}$, hence $a \gg b$, hence Unanimity is satisfied.

The social ranking of a versus b under $F_{\mathcal{U}}$ depends solely on the pairwise comparisons, the set $\{i : a >^i b\}$, which is determined solely by voters' individual rankings of a versus b , independently of all other alternatives, hence IIA holds. □

Proposition 4.13. *A voter $d \in V$ is a dictator for $F_{\mathcal{U}}$ if and only if $\mathcal{U} = \mathcal{U}_d$.*

Proof. ($\Rightarrow$) Assume $d \in V$ is a dictator. Then for any $a \neq b$ and any profile P , $a >^d b \iff a \gg b$. That would suggest that $\{i : a >^i b\} \in \mathcal{U} \iff d \in \{i : a >^i b\}$, which exactly implies $\mathcal{U} = \mathcal{U}_d$.

($\Leftarrow$) If $\mathcal{U} = \mathcal{U}_d$,

$$\begin{aligned} a \gg b &\iff \{i : a >^i b\} \in \mathcal{U}_d \\ &\iff d \in \{i : a >^i b\} \\ &\iff a >^d b \end{aligned}$$

thus d is a dictator. □

4.4.2 Decisive Coalitions give Ultrafilters

We now work in the opposite direction. Given any social welfare function F satisfying Unanimity and IIA, we show that its collection of decisive coalitions $\mathcal{U}_F$ necessarily carries the structure of an ultrafilter on the voters, V .

Definition 4.14 (Decisive coalition). Let $F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$ satisfy Unanimity and IIA. Coalition $S \subseteq V$ is decisive for F if, for every $P = (>^1, >^2, \dots, >^n)$ and every pair $a \neq b$ in $\mathcal{A}$,

$$(\forall i \in S, a >^i b) \implies a \gg b,$$

independently of how voters outside S rank a and b . The decisive coalition of F is

$$\mathcal{U}_F := \{S \subseteq V : S \text{ is decisive for } F\}.$$

Remark 4.15. Note that $d \in V$ is a dictator for F if and only if $\{d\} \in \mathcal{U}_F$.

Theorem 4.16 (Decisive Coalitions form Ultrafilters). *Let $|\mathcal{A}| \geq 3$, and suppose $F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$ satisfies Unanimity and Independence of Irrelevant Alternatives. Then $\mathcal{U}_F$ is an ultrafilter on V .*

Proof. We check that $\mathcal{U}_F$ satisfies all of the axioms for filter and the additional ultrafilter condition.

- (1) By Unanimity, $V \in \mathcal{U}_F$ always, and $\emptyset \notin \mathcal{U}_F$ since the empty coalition cannot force any social outcome, these properties are implied in Propositions 3.23 and 3.22.
- (2) Suppose $S, T \in \mathcal{U}_F$. An argument analogous to Lemma 3.24 shows that $S \cap T \in \mathcal{U}_F$. Pick any distinct $a, b, c \in \mathcal{A}$. Since S is decisive, $a \gg b$ whenever all of S prefers a to b . Since T is decisive, $b \gg c$ whenever all of T prefers b to c . By transitivity, $a \gg c$. Since only voters in $S \cap T$ held both preferences, $S \cap T$ is decisive for (a, c) , hence $S \cap T \in \mathcal{U}_F$.
- (3) If $S \in \mathcal{U}_F$ and $S \subseteq T$, then whenever every voter in T ranks $a > b$, then every voter in S does so as well. Decisiveness of S forces $a \gg_F b$, hence $T \in \mathcal{U}_F$.

Finally, for every $S \subseteq V$, exactly one of S or S^c must lie in $\mathcal{U}_F$. We can prove that at least one of the two is in $\mathcal{U}_F$. Fix $a, b \in \mathcal{A}$, $a \neq b$. Consider the profile where every voter in $S \subseteq V$ ranks $a > b$ and every voter in $S^c \subseteq V$ ranks $b > a$. Since our function gives a strict linear ordering $\gg_F$, either $a \gg_F b$ or $b \gg_F a$ will hold.

If $a \gg_F b$, by Independence of Irrelevant Alternatives, this social ranking will have depended only on the voters which preferred a to b , which were exactly S in this profile. Hence, if whenever all of S prefers a to b we get $a \gg_F b$, it must mean that S was decisive hence $S \in \mathcal{U}_F$. Alternatively if $b \gg_F a$, the alternate argument gives $S^c \in \mathcal{U}_F$. $\square$

4.4.3 Arrow's Theorem

The constructions in Sections 4.4.1 and 4.4.2 allow us to argue that there exists a one-to-one correspondence between ultrafilters on V and fair social welfare functions for the elections with V voters and $\mathcal{A}$ alternatives.

Theorem 4.17. *For $|\mathcal{A}| \geq 3$, the map $\mathcal{U} \mapsto F_{\mathcal{U}}$ is a one-to-one mapping from the set of ultrafilters on V to the set of fair social welfare functions for elections with V voters and $\mathcal{A}$ alternatives, with inverse $F_{\mathcal{U}} \mapsto \mathcal{U}$.*

Proof. By Theorem 4.12, each ultrafilter gives us a social aggregation function that satisfies the required conditions. By Theorem 4.16, any decisive coalition from a function satisfying Unanimity and IIA must be an ultrafilter. $\square$

Theorem 4.18. *Arrow's Impossibility Theorem Let $|\mathcal{A}| \geq 3$ and $|V| \geq 2$. If $F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$ satisfies Unanimity and Independence of Irrelevant Alternatives, then F is a **dictatorship**.*

Proof. By Theorem 4.17, there exists an ultrafilter $\mathcal{U}$ on V such that $F = F_{\mathcal{U}}$. Since V is finite, by Proposition 4.10 it must be the case that $\mathcal{U} = \mathcal{U}_d$ for some unique $d \in V$. By Proposition 4.13, d is a dictator. $\square$

4.5 Infinite Voters

The proof of Arrow's Theorem via ultrafilters allows us to better understand where all of our systems break down. It is necessarily the fact that, because we have a finite number of voters, any decisive coalition we find is a principal ultrafilter and therefore a dictator.

Under an infinite electorate, however, we can construct fair, dictator-free social welfare functions. The argument follows.

Definition 4.19 (Non-principal Ultrafilters). Ultrafilters $\mathcal{U}$ on V are non-principal if they contain no finite set: for every finite $S \subsetneq V$, $S \notin \mathcal{U}$. Alternatively, $\mathcal{U}$ is non-principal if $\mathcal{U} \neq \mathcal{U}_d$ for any $d \in V$.

By Proposition 4.10, non-principal ultrafilters cannot exist on a finite set. We now show they are guaranteed to exist on any infinite set — as we will note below, this existence is irrevocably tied to the belief in the Axiom of Choice.

Definition 4.20 (Fréchet Filter). Let V be infinite. The *Fréchet Filter*, also known as the *Cofinite Filter* on V , is:

$$\mathcal{F} := \{A \subseteq V \mid V \setminus A \text{ is finite}\}$$

Intuitively, a coalition “large” in $\mathcal{F}$ if and only if it omits finitely many numbers.

Proposition 4.21. *The Fréchet Filter on an infinite set V is a filter, but not an ultrafilter.*

Proof. We verify that this set satisfies the filter axioms taking it with respect to $V = \mathbb{N}$ for concreteness.

- (1) $\mathbb{N} \setminus \mathbb{N} = \emptyset$ is finite, hence $\mathbb{N} \in \mathcal{F}$. On the other hand, the complement of $\emptyset$ has an infinite complement, all of $\mathbb{N}$, hence $\emptyset \notin \mathcal{F}$.
- (2) If $A \in \mathcal{U}$ and $A \subseteq B$, then $\mathbb{N} \setminus B \subseteq \mathbb{N} \setminus A$, which is finite, so $B \in \mathcal{U}$.
- (3) If $A, B \in \mathcal{U}$, $A \cap B = \mathbb{N} \setminus (\mathbb{N} \setminus (A \cap B)) = (\mathbb{N} \setminus A) \cup (\mathbb{N} \setminus B)$ is a finite union of finite sets, hence finite, therefore $A \cap B \in \mathcal{U}$.

However, notice that $\mathcal{F}$ is not an ultrafilter. Consider $\text{Even} = \{2, 4, \dots\}$ and $\text{Odd} = \{1, 3, \dots\}$. Both are infinite and are the complement of each other, so neither has a finite complement, meaning neither belong to $\mathcal{F}$. Since the ultrafilters need exactly one S or S^c to belong to $\mathcal{F}$, this is a violation and the Fréchet Filter is not an ultrafilter. $\square$

Proposition 4.22. *The Fréchet filter $\mathcal{F}$ is not contained in any principal ultrafilter $\mathcal{U}_d$.*

Proof. Suppose for the sake of contradiction that $\mathcal{F} \subseteq \mathcal{U}_d$ for some $d \in V$. Consider the cofinite set $S = V \setminus \{d\}$. Then, $S \in \mathcal{F} \subseteq \mathcal{U}_d$. But $d \notin S$, which contradicts the definition of $\mathcal{U}_d$. $\square$

We now use Zorn's lemma to extend $\mathcal{F}$ to a ultrafilter.

Lemma 4.23 (Zorn's Lemma, 1935). *If every chain in a partially ordered set contains an upper bound, then the set has at least one maximal element.*

Remark 4.24. Zorn's Lemma is equivalent to the Axiom of Choice. If one does not accept the Axiom of Choice, the following non-principal ultrafilter existence argument breaks down entirely. In fact, if following Zermelo-Fraenkel set theory which does not necessarily subscribe to the axiom of choice, it is indeed consistent that non-principal ultrafilters do not exist.

Theorem 4.25 (Existence of Non-Principal Ultrafilters). *On any infinite set V , there exists a non-principal ultrafilter.*

Proof. Let $\mathcal{F}$ be the Fréchet filter on V . Consider the following set:

$$\mathcal{F}^* := \{\mathcal{G} : \mathcal{G} \text{ is a filter on } V \text{ and } \mathcal{F} \subseteq \mathcal{G}\}$$

ordered by inclusion. $\mathcal{F}^*$ is non-empty since $\mathcal{F} \in \mathcal{F}^*$.

For any chain $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \dots$ in $\mathcal{F}^*$, the union $\bigcup_i \mathcal{G}_i$ is a filter containing $\mathcal{F}$, hence an upper bound in $\mathcal{F}^*$.

By Zorn's Lemma, $\mathcal{F}^*$ contains a maximal element $\mathcal{U}$. By Proposition 4.5, $\mathcal{U}$ must be an ultrafilter.

Since $\mathcal{F} \subseteq \mathcal{F}^* \subseteq \mathcal{U}$, and because by Proposition 4.22 $\mathcal{F}$ cannot be contained in any principal ultrafilter, it follows that $\mathcal{U}$ must be non-principal! $\square$

Corollary 4.26. *On an infinite electorate V , the social welfare function $F_{\mathcal{U}}$ defined by any non-principal ultrafilter $\mathcal{U}$ is fair and has no dictator.*

Proof. Fairness follows from Theorem 4.12. By Proposition 4.13, d is a dictator for $F_{\mathcal{U}}$ if and only if $\mathcal{U} = \mathcal{U}_d$. Since $\mathcal{U}$ is non-principal, no such d exists, therefore on an infinite electorate, there exists a non-dictatorial *fair* system. $\square$

Chapter 5

Navigating Impossibility: Democracy in the 21st Century

The key result presented in this paper is the impossible nature of fair democratic aggregation. Chapters 3 and 4 demonstrate that any social welfare function satisfying a very minimal set of fairness axioms must be dictatorial — a conclusion that feels completely antithetical to the democratic project many countries manifest. Regardless, the world keeps governing itself and must keep doing so. The question remains... how should we go about treating democracy, elections and social aggregation?

5.1 Current State of the World

In practice, only a handful of electoral mechanisms are used worldwide. Figures 5.1 and 5.2 situate these mechanisms in their contemporary contexts, illustrating their geographic distribution and mapping the systems used in over 170 countries alongside their *Freedom House* scores. This index measures political rights and civil liberties, awarding up to 100 points based on indicators such as free elections, pluralism, and freedom of expression.

Together, these maps reveal a striking pattern: no electoral mechanism can guarantee individual freedom. The variance in the freedom scores within each system family far exceeds the variance between them. Plurality democracies range from Canada (97) to Turkmenistan (1); proportional systems span Finland (100) to Equatorial Guinea (5). The choice of aggregation mechanism, seems to tell us quite little about the freedom a countries' citizens

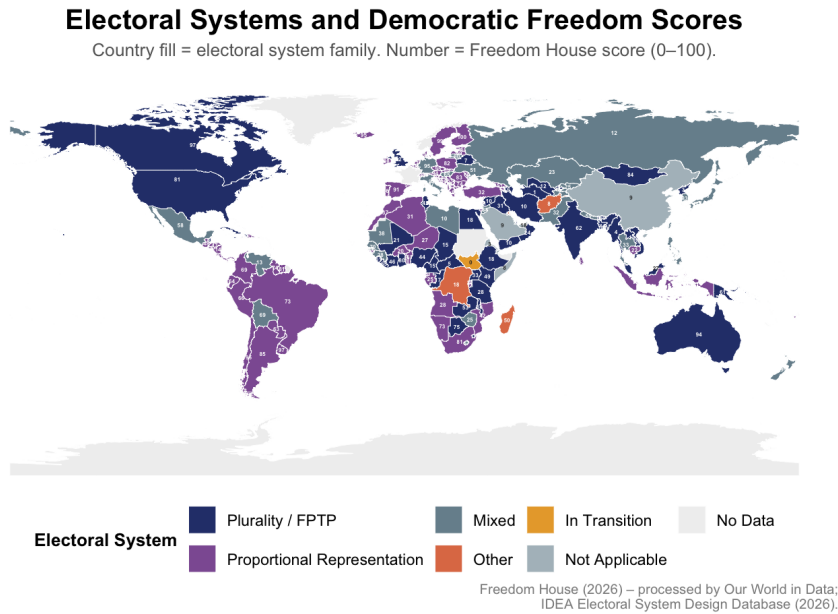


Figure 5.1

enjoy. While we cannot necessarily measure the level of electoral satisfaction experienced in these countries, this empirical picture suggests that something beyond voting mechanisms determines democratic health. Arrow’s theorem already tells us that all mechanisms are fatally flawed. On top of this, data seems to suggest that in practice: institutions, history, culture, economic development, and civic participation matter far more than whether a country aggregates votes by Plurality or Proportional Representation.

5.2 Philosophical Exploration of What to do Under Impossibility

The proof of Arrow’s Impossibility Theorem via Ultrafilters in Chapter 4 was able to formulate the mathematical reason why dictatorship is forced on finite electorates, and in doing so, allowed us to understand the unique condition under which it can be avoided. Ultimately, only infinite sets offer non-principal ultrafilters. It follows that only social welfare functions on infinite electorates can avoid dictatorial outcomes while satisfying Unanimity

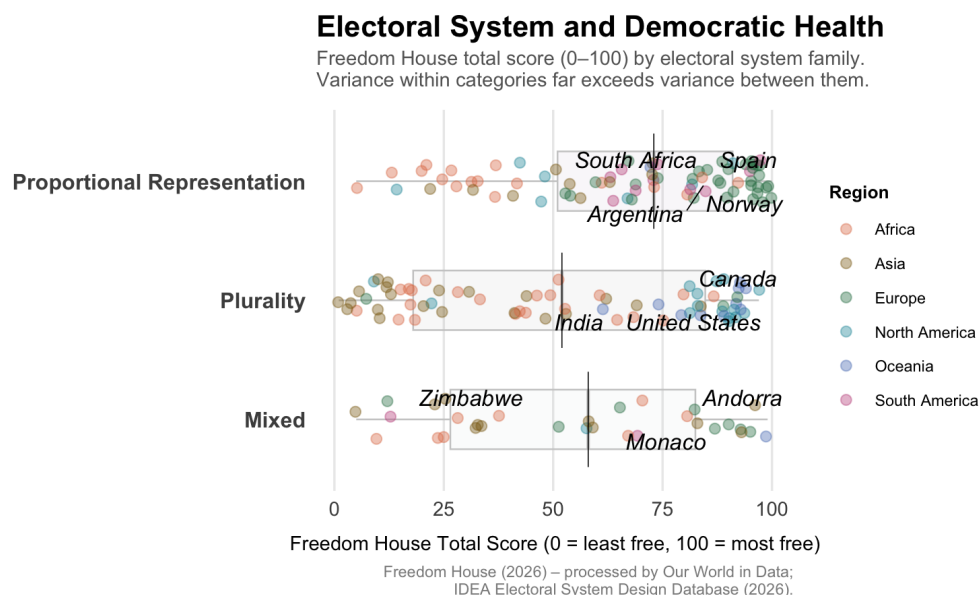


Figure 5.2

and IIA.

Unfortunately, this result does not get us much further. Even if we modelled society as an infinite electorate Arrow’s impossibility would fail to dissolve. Kirman and Sondermann explored this exact narrative, finding that even when the electorate is “infinite” and non-principal ultrafilter generates the social welfare function, something deserving of the title of dictatorship remains Kirman and Sondermann (1972). The free ultrafilter, the decisive coalition structure underlying the social welfare function in the infinite scenario still corresponds to an arbitrarily small group determining social outcomes. For any positive proportion of the electorate, we could show that no matter the size, we could find a smaller deciding group deciding the social ordering.

Unfortunately, modelling society as the continuity of generations across time leading to an infinite electorate does not dissolve the concentration of power, it merely diffuses it into a non-identifiable elite coalition. This new invisible dictator cannot be pointed to at all.

If changing our social aggregation function or expanding the electorate infinitely cannot get us any further in tackling these questions, we must look

this problem from a different point of view. The only thing left to change is the input to the function.

Arrow's theorem necessarily operates on a universal domain, but it is this assumption, as much as our two fairness conditions that drives the impossibility of the question we are trying to solve. Domain restrictions can potentially get us closer to satisfactory solutions.

The main result in this line of argument was already explored in Chapter 2: single-peakedness. Black's Median Voter Theorem proves that when individual preferences are single peaked with respect to a common ordering of alternatives, we can come up with solutions that we like. The Condorcet winner always exists and equals the preference of the median voter. Furthermore, majority rule works perfect under this condition. Arrow's impossibility, therefore, can be broken down by restricting the domain on which our function operates.

The practical objection, as aforementioned, is the likelihood of this assumption. We need to ensure that we have prior consensus on an ideological dimension — a shared ordering of the alternatives. This assumption may be unrealistic, especially when considering the complex multi-dimensional nature of human thought. So much as disagreeing on the fundamental framing of question makes satisfying this restriction near impossible.

List et al. (2013) argue that not all is lost. They provide an important empirical finding: single-peakedness may be generated through deliberation. Using data from Deliberative Polls, they tested whether deliberation moved preference profiles closer to single-peakedness by measuring the proportion of individuals whose preferences are single-peaked with respect to the question's main dimensions.

Their work showed that deliberation increased proximity to single-peakedness. Deliberation was not necessarily leading to consensus, but rather was inducing *meta-agreement*: a shared understanding of the dimension along which the question was to be evaluated, without the requirement of agreeing on the answer.

Arrow's theorem requires that we fuse together whatever individual preferences individuals happen to hold even if they are incoherent in their collective structure. Single-peakedness doesn't require everyone to agree, it just requires that they conceptualise the question being voted on along a single, same dimension. What deliberation produces is agreement about the terrain of the question, not about the answer. Meta-agreement allows participants to come to recognise a common semantic dimension.

The practical implications of deliberation, as presented by List et al. (2013) is that it can move use towards the domain restriction under which Black's theorem allows majority rule to work perfectly. If democratic societies invest in figuring out the conditions needed for genuine conversation, they may be able to reduce the likelihood of obtaining Condorcet cycles, allowing majority rule to function as intended.

5.3 Closing Thoughts

This body of work has focussed on a question that mathematics declares impossible. To close on impossibility alone, however, would be to miss something essential about what this result implies in a contrapositive manner.

Arrow did not find a flaw in democracy. He found a flaw in the idea that human disagreement can be perfectly resolved in formal procedure.

And yet.

Humanity has governed itself for millennia, building civilisations, crossing oceans, electing respectable presidents and landing on the moon. None of these accomplishments were mathematically guaranteed. None of these results emerged from a procedure satisfying Arrow's axioms. They grew from something the axioms were never designed to capture: the human capacity for dialogue, compromise, and for the kind of collective imagination that allows people to coexist in disagreement.

It is worth taking this reflection in for a moment. The first fire was not lit by a single individual imposing their will on the rest — it was lit by people watching one another, imitating, experimenting and communicating across the gaps of human experience. Every step that has followed — from the wheel; to the printing press; to vaccines; to microchips — has had the same origin. Not the mechanical aggregation of fixed preferences, but the frustrating process of people changing their minds together.

Arrow's model, by mathematical necessity, removes this layer from the equation. The model assumes preferences are fixed before aggregation begins; that each voter arrives at the ballot with a fully formed ranking, disregarding what everyone else has said, thought or discovered. This is a useful mathematical reduction. It is not, however, a description of how human beings make decisions. In reality, the preferences we bring to collective decisions are shaped by every conversation we have had, every argument we have lost, every perspective that has surprised us into reconsidering our own. The inputs

to the aggregation function are not fixed. They are living and ever-updating.

These final words are not refuting Arrow's work. They only suggest that the democratic project, when properly understood, is not only about finding the correct aggregation mechanism. Fostering democracy is about cultivating the conditions under which people can engage in civil discourse: creating an environment where we can talk honestly to each other and sit in disagreement without retreating into hostility and isolation.

The great democratic achievements of the twentieth century were not made by finding a better voting rule. They were won by people who changed the terms of the conversation: who reframed questions, expanded who was included in the deliberation, and appealed to values that their opponents already held but had not yet applied consistently. Nelson Mandela did not discover a new social welfare function. He changed the preference profile of a nation.

For all of its precision, Arrow's theorem cannot capture that the deepest response to the impossibility of perfect aggregation is not a better procedure, but a better question. Far from telling us that democracy is futile, it tells us, with mathematical precision, that our focus should be toward everything that must happen before any ballot is ever cast.

Appendix A

Laplace & Borda Count

In this section, we aim to show that linear positional weights (as given by Borda Count) are the unique optimal estimator of candidate merit (and therefore ranking) when we assume that the candidates' merit is uniformly distributed. This argument appears primarily in *Théorie Analytique des Probabilités* (Laplace, 1820) and the *Essai philosophique sur les probabilités* (Laplace, 1814), both by Pierre-Simon Laplace. It was, in fact, one of the earliest applications of Maximum Likelihood Estimation to social choice theory.

Definition A.1 (Positional Scoring). Consider a set of k candidates. A positional scoring rule can be defined by $\vec{w} = (w_1, w_2, \dots, w_k) \in \mathbb{R}^k$ such that $w_1 \geq w_2 \geq \dots \geq w_n$, and $w_1 > w_n$. A voter ranking a candidate to a i -th position assigns them a score of w_i .

Example A.2. One such example of a positional scoring rule is Borda Count, where:

$$w_i = k - i$$

Definition A.3 (Latent Merit). Assume each candidate $c \in [k]$ has an inherent, true unobserved merit $\mathcal{V}_c$ (*valeur de mérite*). We assume that for any voter, these merits are independent and identically distributed random variables from a standard uniform distribution:

$$\mathcal{V}_c \sim U(0, 1)$$

We assume that voters strictly rank candidates in descending order of their perceived $\mathcal{V}_c$, so that the most highly valued candidate receives rank 1.

Definition A.4 (Order Statistic). The value of a sample of random variables arranged in ascending order of magnitude, which here we denote as:

$$\mathcal{V}_{(1)} \leq \mathcal{V}_{(2)} \leq \dots \leq \mathcal{V}_{(k)}$$

They represent the sorted observations with $\mathcal{V}_{(1)}$ being the smallest observation, and $\mathcal{V}_{(n)}$ being the largest.

Lemma A.5 (Expected Value of Uniform Order Statistics). *Let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be an i.i.d. sample from $U(0, 1)$, with associated order statistics $\mathcal{V}_{(1)} \leq \dots \leq \mathcal{V}_{(k)}$. If a candidate is ranked in the i -th position from the top, their expected latent merit is:*

$$E[\mathcal{V} \mid \text{Rank} = i] = \frac{k - i + 1}{k + 1}$$

Proof. This follows from a standard statistical result. A candidate ranked i -th from the top corresponds to the $(k - i + 1)$ -th smallest value among the k latent merits, that is, the order statistic $\mathcal{V}_{(k-i+1)}$.

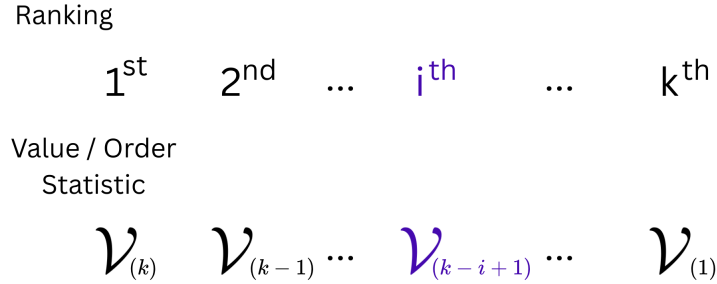


Figure A.1: Ranking vs Order Statistic

For an i.i.d. sample of size k from $U(0, 1)$, the r -th order statistic follows a Beta distribution:

$$\mathcal{V}_{(r)} \sim \text{Beta}(r, k - r + 1)$$

with expectation $\frac{r}{k+1}$. Setting $r = k - i + 1$ gives:

$$E[\mathcal{V} \mid \text{Rank} = i] = E[\mathcal{V}_{(k-i+1)}] = \frac{k - i + 1}{k + 1}$$

□

Proposition A.6 (Optimal Estimator Minimising MSE). *Let $\mathcal{V}$ denote the true merit of a candidate and R their observed rank. Among all weighting vectors $\vec{w}$, the one minimising the mean squared error $\mathcal{L} = E[(\mathcal{V} - \langle \vec{w}, R \rangle)^2]$ assigns to rank i the weight:*

$$w_i^* = E[\mathcal{V} \mid R = i]$$

Proof. This is the classical result that the conditional expectation $E[\mathcal{V} \mid R]$ is the minimum mean squared error estimator of $\mathcal{V}$ given R . $\square$

Theorem A.7 (Borda Count Optimality). *Under the latent merit model (Definition A.3), the positional scoring rule (Definition A.1) that minimises the MSE between assigned weights and candidates' true latent merits is equivalent to Borda Count.*

Proof. By Proposition A.6, the MSE-minimising weights are the conditional expectations of latent merit given rank. By Lemma A.5, for a candidate ranked i -th from the top:

$$w^*(i) = \frac{k - i + 1}{k + 1}$$

For each candidate c , summing these weights across all n voters gives a total score:

$$S_c = \sum_{j=1}^n w^*(R_{cj}) = \frac{1}{k+1} \sum_{j=1}^n (k - R_{cj} + 1)$$

where R_{cj} denotes the rank assigned to candidate c by voter j . Since $\frac{1}{k+1}$ is a positive scaling factor and $\sum_{j=1}^n \frac{1}{k+1}$ is a constant added equally to every candidate's score, neither affects the relative ordering of candidates. The ranking-equivalent score reduces to:

$$\tilde{S}_c = \sum_{j=1}^n (k - R_{cj})$$

This is precisely the Borda Count, where candidate c receives $k - R_{cj}$ points from voter j . Hence the unique optimal positional scoring rule under the uniform latent merit model is Borda Count. $\square$

Table of Notation

Symbol	Description
<i>Sets & Elections</i>	
$\mathcal{A}$	Finite set of alternatives $\{a_1, a_2, \dots, a_k\}$, with $ \mathcal{A} \geq 3$
V	Finite set of voters $\{1, 2, \dots, n\}$, with $n \geq 2$
k	Number of alternatives, $k = \mathcal{A} $
n	Number of voters, $n = V $
<i>Preferences & Profiles</i>	
$\succ^i$	Strict preference ordering of voter i
$\mathcal{B}_{\mathcal{A}}$	Set of all strict linear orders on $\mathcal{A}$
$P = (\succ^1, \dots, \succ^n)$	Preference profile; an n -tuple of individual ballots
$\succcurlyeq$	Societal preference order output by F
$x \succcurlyeq y$	Society prefers x to y
$P^{a \succ b}$	Coalition of voters preferring a to b under profile P
<i>Social Welfare Functions</i>	
F	Social welfare function $F : \mathcal{B}_{\mathcal{A}}^n \rightarrow \mathcal{B}_{\mathcal{A}}$
$F_{\mathcal{U}}$	Social welfare function induced by ultrafilter $\mathcal{U}$
<i>Voting Rules</i>	
$S(x)$	Plurality or Borda score of alternative x
$\text{Llull}(x)$	Llull score of alternative x
$\text{Dod}(x)$	Dodgson score of alternative x

Symbol	Description
Maj	Majority threshold $\lfloor n/2 \rfloor + 1$
$\mathcal{A}^{(r)}$	Remaining alternatives in round r of STV or Nanson
$S^{(r)}(x)$	Vote count for alternative x in round r
<i>Decisiveness</i>	
$\mathcal{D}_{a>b}$	Collection of coalitions weakly decisive for a over b
$\mathcal{D}^*_{a>b}$	Collection of coalitions strongly decisive for a over b
$\mathcal{D}$	Shorthand for $\mathcal{D}_{a>b}$ (independent of any pair of alternatives, by Lemma 3.26)
M	Minimal decisive set
<i>Filters & Ultrafilters</i>	
$\mathcal{P}(V)$	Power set of V
$\mathcal{F}$	Filter on V
$\mathcal{U}$	Ultrafilter on V
$\mathcal{U}_d$	Principal ultrafilter generated by voter $d : \{S \subseteq V : d \in S\}$
$\mathcal{U}_F$	Collection of decisive coalitions of F
$\mathcal{F}$	Fréchet (cofinite) filter on an infinite set
<i>Appendix: Latent Merit Model</i>	
$\mathcal{V}_c$	Latent merit of candidate c , drawn from $U(0, 1)$
$\mathcal{V}_{(r)}$	r -th order statistic of the latent merits

Symbol	Description
R_{cj}	Rank assigned to candidate c by voter j
$\vec{w}$	Positional scoring vector $(w_1, \dots, w_k)$
$w^*(i)$	MSE-optimal weight for rank position i
S_c	Total score of candidate c
$\tilde{S}_c$	Ranking-equivalent score of candidate c

Acknowledgments

This body of work would not have been possible without the ceaseless help of my thesis advisor, Professor de Silva, who has shown exceptional patience throughout the entire process. His guidance has allowed me to enjoy every minute of what has felt like a tangential mathematical exploration, and his mentorship has been instrumental in my development as both a presenter and as a communicator.

I would also like to express my profound indebtedness to my statistical mentors, Professor Jo Hardin and Professor Gabriel Chandler. Their unwavering support and encouragement have allowed me to develop a true passion for statistics, and have solidified my own belief in my own mathematical abilities. I am on the journey I am today largely because of their faith in me. Furthermore, I would like to thank the faculty of the Pomona College Department of Mathematics and Statistics, each of whom has contributed to my academic growth in distinct and lasting ways.

I also owe a special debt of gratitude to Ms. Leon, my middle school and high schoolteacher, whose instruction first sparked my interest in this field, and resulted in me enrolling in Professor Shahriari's Linear Algebra my first semester at Pomona.

Finally, I am eternally grateful to my parents and siblings for their constant encouragement to pursue my ambitions, for instilling in me a faith in God, and for providing a foundation of unwavering support. My deepest thanks also go to my friends at Pomona.

To me, Pomona has never been defined by its buildings or its curriculum, but by the community and the shared journeys that have shaped the person I am today.

Technology Statement

Used Grammarly to clean up grammatical faults.

Used R to create plots and analyse data.

Used Gemini and Claude Deep Research mode to find papers and sources, as well as relevant information. Used NotebookLM to parse and read through papers.

Used the aid Gemini Pro 3 to create some tikz-cd diagrams, and to clean up some notation, as well as the occasional tinkering of verbose phrases. Used Claude.ai to help check for typos and to generate the table of notation after all the work was finished.

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